

Strategic Multinational Location Choice with Endogenous Pricing*

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Abstract

We study strategic multinational location choice with endogenous pricing. In this combinatorial discrete choice problem, a location's net return can switch from negative to positive and back as the firm's network expands. Our profit bounds link market benefits and location costs to the same portfolio, allowing globally optimal best responses to given rival strategies. We apply the framework to a calibrated model of cloud providers serving customers across borders. Requiring EU public-sector customers to be served from within the EU expands the modeled providers' EU networks relative to allowing supply from any country. Requiring supply from each customer's own country instead contracts them. Moving from EU-wide to national sourcing also raises prices for EU private-sector customers and customers outside the EU as providers adjust their networks. In China, higher costs of serving customers from abroad increase Chinese providers' domestic presence but shrink their worldwide networks and reduce their operating profits. Network adjustment also raises prices outside China. In both experiments, firms' responses to competitors' changing networks substantially moderate worldwide network contraction while leaving price increases for directly affected customers largely unchanged.

Keywords: multinational production, location choice, portfolio equilibrium, variable markups, cloud computing, digital trade

JEL Codes: F12, F23, L86, C72, C62, L13

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1. INTRODUCTION

A restriction on market access can change the customers a facility can serve and, with them, the value of locations throughout a multinational network. A firm's response depends on its other facilities and on how competitors reorganize their networks. The country imposing the restriction need not be the country where networks shrink or customers pay more. We study these interdependent location choices when the return to a location need not rise or fall consistently as networks expand. The question is how to solve for complete networks and what firms' strategic responses imply for the international effects of market-access restrictions.

Cloud computing makes this problem concrete. Concerns about digital sovereignty are being translated into requirements on the provision and procurement of cloud services. The European Commission's cloud procurement framework and Australia's hosting certification scheme incorporate sovereignty and control (European Commission, 2026a; Australian Government, 2021).¹ Providers serve customers across borders from core regions, which are groups of data centers, and from smaller local facilities. AWS's Copenhagen Local Zone, for example, is linked to a parent region in Stockholm. A provider can thus supply a country without locating a core region there, while a core abroad can support service in several markets. A geographic restriction changes the customers these facilities can serve and the profitability of other locations in the same network. Rival providers face altered market shares and location incentives in turn.

We develop a framework for strategic multinational location choice with endogenous pricing. Firms observe private location costs and choose complete portfolios before competing in customer markets. A portfolio determines delivered costs and market access, and the resulting prices and shares determine its operating profits. Firms choose both the number and the composition of their locations. The cloud application combines multinational production with the Atkeson–Burstein (AB) share-dependent markup schedule.

Two distinct difficulties arise. Even holding rivals' portfolios fixed, the best network cannot generally be assembled by ranking individual locations. Adding facilities changes costs, prices, and market shares, and can first raise and then lower the return to another

¹Digital sovereignty encompasses several claims about authority over data and technology (Pohle and Thiel, 2020). Our applications study the location and competition effects of geographic supply restrictions, holding legal ownership and operating arrangements fixed.

facility. Across firms, the network that is optimal against current rivals need not remain optimal once they respond. Unique prices at a given collection of networks resolve neither difficulty: own-network optimization and equilibrium in portfolio choices are separate problems.

The cloud model illustrates why location returns can reverse sign. Consider a provider serving European customers from a small network. An additional core may attract little business if its service remains uncompetitive in cost and accessibility. Once other cores have improved those attributes, the same addition can attract more customers and become worth its location cost. But when the provider already serves a large share of the market, there is less scope to gain customers, and the additional profit can again fall below that cost. A core can therefore complement other locations at one network size and substitute for them at another. Under variable markups, rival composition and repricing also affect these returns. We characterize these sign reversals and show that they can also arise under constant markups.

We construct profit bounds for the search over all feasible portfolios. Market benefits and location costs must arise from the same portfolio. A loose bound can combine one country's advantage in Europe, another's advantage in Asia, and a third country's low cost, even when only one facility is allowed. Our bounds restrict these infeasible combinations by retaining the same firm's core and service choices across markets, including the parents needed by local facilities. They allow endogenous portfolio size even when a location's net return changes sign more than once as the network expands.

These *completion bounds* limit the best profit attainable by a group of unexamined networks. If the bound falls below the profit of a feasible network already found, the whole group can be discarded. Branching resolves remaining groups, and unresolved bounds measure the scope for further improvement. With 20 optional countries and rivals held fixed, the method finds the same optimal profits as exhaustive enumeration a median 235 times faster, including preparation (Figure 2).

To compute an equilibrium, we aggregate private-cost types' network choices into portfolio distributions and update those distributions across firms. Searches over all feasible portfolios assess profitable deviations at the resulting profile; consistency across firms is a separate requirement from optimality against fixed rivals. Additional sufficient conditions relate equilibrium uniqueness to private-cost dispersion and strategic responses.

The applications quantify why strategic portfolio responses matter. Eight cloud providers choose core presence across 42 countries, with other suppliers represented in aggregate. Two experiments alter the geographic reach of service supply. The EU experiment compares EU-wide eligibility with domestic eligibility for the same regulated customers, relative to a baseline without these geographic eligibility restrictions.² The China experiment lowers or raises offshore delivery costs around existing restrictions on directly controlled mainland cores and partner-operated services. Its baseline therefore already represents a segmented market.

We calibrate each provider’s net location-cost level to the number of countries in which it has a core presence. Worldwide market-share targets discipline providers’ relative competitiveness and the aggregate contribution of other suppliers. These parameters are calibrated jointly because network choices and market shares affect one another. Elasticities, spatial cost coefficients, service connections, and private-cost dispersion are assigned separately. Regional composition is assessed as a model-fit diagnostic rather than an additional matching target. The calibrated parameters remain fixed across policy comparisons.

In each experiment, we separate the immediate effect at unchanged networks from portfolio adjustment against unchanged rivals and from adjustment when rivals respond as well. Prices are jointly determined in every comparison. Competitor responses leave much of the directly affected price increase intact but substantially moderate worldwide network contraction. In the EU comparison, the global contraction is about one-third smaller once rivals respond; in the larger China tightening, it is almost one-half smaller. The locations and providers that adjust also determine price effects in markets whose own access rules have not changed. A similar regulated-market price effect can therefore conceal a different international adjustment.

The applications also distinguish domestic presence from the global network supporting it. EU-wide eligibility directs regulated demand toward EU origins while preserving access to customers throughout the participating countries. It expands the modeled providers’ aggregate EU core presence relative to the baseline. National eligibility also fa-

²The Commission’s 2026 proposal for a Cloud and AI Development Act includes an EU-wide framework for assessing cloud sovereignty (European Commission, 2026c); member states also apply their own public-sector qualifications, such as France’s SecNumCloud (Agence nationale de la sécurité des systèmes d’information, 2022). The experiment isolates the geographic scope of eligible service supply. It does not reproduce ownership requirements or the full qualifications in either the proposal or existing national schemes.

vors domestic origins, but each can serve only domestic regulated demand; that demand need not justify a core. National eligibility reduces even aggregate EU core presence relative to the baseline. In China, higher offshore costs increase Chinese providers' probabilities of operating a mainland core while reducing their worldwide networks and operating profits. Those firms also supply mainland customers from offshore facilities, so the restriction raises costs within their own networks. Greater domestic presence need not translate into higher profits.

The paper builds on quantitative trade models in which choices across countries are interdependent. Earlier destinations can change later entry costs (Morales et al., 2019; Alfaro-Ureña et al., 2025), sourcing locations jointly determine input costs (Antràs et al., 2017), and production locations jointly determine market access (Tintelnot, 2017; Castro-Vincenzi, 2024; Antràs et al., 2026). These models motivate treating the complete location set as the object of choice. Our focus is a game in which the value of that set depends on competitors' networks and on downstream pricing. This connects multinational location choice to strategic interaction across markets (Bulow et al., 1985) and to models of product and network choice (Holmes, 2011; Wollmann, 2018; Fan and Yang, 2020; Nishida, 2015).

Methods for combinatorial choice exploit restrictions on marginal returns to avoid evaluating every combination. Jia (2008) uses complementarity in strategic chain entry, Arkolakis et al. (2026) (AES) develop single-crossing restrictions, and Castro-Vincenzi et al. (2025) use known signs of complementarity and substitutability; related structure appears in fulfillment and plant networks (Houde et al., 2023; Oberfield et al., 2024). AES also branches when its elimination procedure leaves unresolved choices. Sabal (2026) is particularly close in studying strategic portfolio choices by multiple asymmetric firms under private costs. His equilibrium algorithm exploits decreasing marginal profits as own and rival product sets expand to bound equilibrium choice distributions, allowing for multiplicity.³ To find the best portfolio at given rival strategies, we instead bound the total profit attainable by a group of networks. The supply and cost structure makes these bounds computable even when marginal location returns change sign in both directions. This allows policy-induced network reorganization to be analyzed while retaining the in-

³Sabal's equilibrium algorithm also extends to the supermodular case, and his estimation bounds require weaker conditions than submodularity. Our sign-reversal example violates both submodularity and supermodularity: a location's net return loses its positive sign both when the network contracts and when it expands.

teractions generated by service technology and pricing, without imposing submodularity or supermodularity on portfolio profits.

The cloud applications connect research on data regulation to strategic multinational production. Existing work studies productivity effects of data restrictions (Ferracane et al., 2020), firms' demand for cloud inputs under privacy law (Demirer et al., 2024), digital-service trade and market access (Sun and Trefler, 2023; Sun, 2024), and competition in cloud services (Manganelli and Schnurr, 2024). Policy research distinguishes restrictions on data transfers from requirements to store or process data locally (Casalini et al., 2021; Del Giovane et al., 2023). We examine how restrictions on the reach of service supply change providers' complete location portfolios and their competitors' choices, transmitting price effects beyond the regulated market.

Sections 2 and 3 present the model and theory. Section 4 describes the network data and calibration, and Section 5 presents the policy experiments and their institutional setting. The appendix contains proofs and supporting evidence.

2. LOCATION PORTFOLIOS AND ENDOGENOUS PRICING

Core regions are groups of data centers supplying cloud services. Smaller local facilities extend those services and may depend on a parent core region, including one abroad.

Providers observe their private location costs, choose core portfolios simultaneously, and then face the downstream pricing system specified below. They know rivals' cost distributions but not their realizations. A portfolio determines the available service paths; regulation determines which paths can serve each customer. These paths determine delivered costs and access, which in turn determine shares, markups, and operating profits. Section 3 develops the portfolio game and solution method for a general finite set of location alternatives.

2.1 Core location, service paths, and regulation

Firm f chooses a set S_f of countries hosting its core infrastructure. Each provider's core regions within a country are grouped into one location alternative, so $K_f = |S_f|$ counts its core-hosting countries. The model captures the extensive margin of core presence across countries; it does not separately choose the number or capacity of regions within a host

country. Aggregate core counts sum these provider-country presences across firms; expected counts average over the private-cost realizations described in Section 2.3 and the portfolio choices they induce.

Installed local facilities can provide service from a country without a core there. Their locations are held fixed. A local service with a documented parent operates when that parent core is in the provider’s chosen portfolio, including when the parent is abroad. When the parent is unconfirmed, we maintain observed local availability independently of the chosen cores.⁴ Several local facilities in the same country provide one service origin.

A customer segment n is a destination country and customer group. The EU application distinguishes regulated public demand from private demand within each participating country; other countries have one customer segment. A regime r determines which supply paths are eligible for each segment.⁵

The service origin is the country in which the modeled core or local service is located. Geographic eligibility applies to that origin. A local service can therefore serve domestic regulated demand when its supporting core is abroad, provided that the required parent core is in the provider’s portfolio. Parent presence and service-origin eligibility are separate conditions: national eligibility does not require the supporting parent to be domestic. For unconfirmed parent links, the maintained local availability is subject to the same service-origin rule.

We represent a provider’s service by a fixed composition of workloads that can be performed with general-purpose virtual machines. As its service quantity expands, workload requirements increase proportionally. Each workload is assigned to its least-cost eligible origin: a larger network can offer a less costly way to perform a given workload, but additional processing of one workload cannot replace another.⁶

Core and local facilities can both provide such compute services, although their product ranges differ. Within this representative workload class, the service mapping distin-

⁴This is an assumption about unobserved support. Section 4.1 describes the local-service data and their parent documentation.

⁵EU public-cloud rules restrict the geographic origin of eligible supply; the China application changes offshore delivery costs while maintaining specified local operating modes. The latter restrictions are detailed in Section 5.2. The policies change which customers a network can reach or the cost of reaching them, and hence the return to its core locations.

⁶A workload includes resources that must be deployed together, such as a virtual machine and its required storage and network functions. Efficient assignment is a maintained assumption; it does not describe automatic migration of customers’ resources between regions. Capacity constraints and congestion are abstracted from.

guishes their technology intensities. Let $u_{fin}^r(S_f)$ indicate that firm f has a core in country i eligible to serve segment n under regime r , and let $o_{fin}^r(S_f)$ indicate coverage through either an eligible core or an operating local path. Both indicators apply the service-origin rule e_{in}^r , which equals one if origin i may serve segment n under regime r and zero otherwise. Section 5.1 specifies this rule for the EU experiment. Parent support determines whether a local path operates before the origin rule is applied. Thus $o_{fin}^r(S_f) \geq u_{fin}^r(S_f)$. Effective supply is

$$t_{fin}^r(S_f) = \underbrace{u_{fin}^r(S_f)}_{\text{core supply}} + \underbrace{\gamma_{\text{loc}}[o_{fin}^r(S_f) - u_{fin}^r(S_f)]}_{\text{local-only supply}}, \quad \gamma_{\text{loc}} = 0.5.$$

A core has intensity one, a local-only path one half, and an unavailable path zero. Core and local supply in the same country are not added together. The relative intensity is assigned, not a measured capacity ratio or the fraction of infrastructure services available locally.⁷

Each eligible origin can lower the cost of serving a customer. Let w_i denote its operating input-cost index, τ_{in} the cost of delivering service from i to segment n , and a_f the provider's relative competitiveness. Delivered marginal cost aggregates effective supply as

$$\Sigma_{fn}^r(S_f) = \sum_i t_{fin}^r(S_f)(w_i \tau_{in})^{-\theta}, \quad c_{fn}^r(S_f) = a_f^{-1}[\Sigma_{fn}^r(S_f)]^{-1/\theta}. \quad (2.1)$$

With independent Fréchet workload efficiency across eligible origins, unit cost is the average of the least-cost ways to perform the fixed workloads. This gives (2.1) after absorbing a finite common factor into a_f . The production-side condition is $\theta > 0$; Appendix D.1 gives the derivation. The parameter θ governs efficiency dispersion across origins, whereas σ below governs substitution across providers. Separating internal aggregation from provider demand follows the distinction in Antràs et al. (2017); Fréchet sourcing also underlies the multinational production model of Tintelnot (2017). A core can improve supply in a country already served by a local facility without creating a new service origin. For illustration, start with a single core origin and hold delivered input

⁷Provider documentation establishes overlap in virtual-machine supply and differences in supported products and features; it does not identify this relative intensity. Appendix E.2 reports the evidence. The weight affects the return to a core in countries already served by local facilities.

costs $w_i \tau_{in}$ equal across origins. At $\theta = 3$, adding a second core origin multiplies marginal cost by $2^{-1/3}$, a reduction of about 21 percent; adding a local-only origin instead multiplies it by $1.5^{-1/3}$, a reduction of about 13 percent. The weight t determines each origin's contribution to this cost reduction.

Customers also value access to a provider's network. Let q_{in} measure accessibility from origin i to segment n . Additional eligible origins can improve access even when the most accessible origin is unchanged:

$$A_{fn}^r(S_f) = \lambda_A^{-1} \log \left(1 + \sum_i o_{fin}^r(S_f) e^{\lambda_A q_{in}} \right), \quad \psi_{fn}^r(S_f) = e^{\chi[A_{fn}^r(S_f) - A_n^{\text{frontier}}] + d_{fn}}. \quad (2.2)$$

Here A_n^{frontier} is access with all origins available, held fixed across regimes so that an eligibility restriction cannot reduce both access and its reference level, offsetting the service loss. The parameter λ_A controls aggregation across origins, χ scales the demand advantage from access, and d_{fn} permits a home-market advantage. The accessibility score is a proxy, not a direct measure of latency or willingness to pay; Section 4.1 describes its construction. Neither a_f nor d_{fn} is separately identified as physical productivity or a causal preference effect. Both cost and access depend on the complete portfolio because local supply can require a parent core elsewhere. A provider with no eligible supply has zero share in that segment.

At a given eligible origin, a core and a local-only service each enter access through $o = 1$, while their cost contributions use $t = 1$ and $t = \gamma_{\text{loc}}$, respectively. Access reflects service geography, whereas cost also reflects supply capability.

2.2 Market shares, variable markups, and pricing

Customer expenditure R_n in each segment is fixed across policy regimes. The real quantity of the service composite, R_n/P_n , therefore falls when its price index rises. The model holds total cloud expenditure and its allocation across customer segments constant. Operating input prices are also fixed: this is a partial-equilibrium model in which core locations

and cloud-service prices adjust.⁸ For each portfolio profile and regime, price-index contributions and revenue shares satisfy

$$y_{fn} = \psi_{fn} p_{fn}^{1-\sigma}, \quad s_{fn} = \frac{y_{fn}}{b_n + \sum_g y_{gn}}, \quad b_n \geq 0. \quad (2.3)$$

The nonnegative term b_n is the aggregate contribution of other suppliers (the residual) to the price index. In the baseline it is common across customer segments, $b_n = b$, with its level calibrated to global shares. It summarizes their available services and prices, rather than physical capacity or their expenditure share. The latter is $b_n/(b_n + \sum_g y_{gn})$ and changes when the eight modeled providers' prices or eligibility change. In the central comparison, these suppliers remain eligible in every segment and their aggregate contribution is fixed. Sensitivity exercises vary their contribution and exposure to restrictions. Residual suppliers do not choose portfolios. A provider without eligible supply contributes zero to the sum.

For a portfolio profile S and regime r , the corresponding price index is

$$P_n^r(S) = \left[b_n^r + \sum_f \psi_{fn}^r(S_f) p_{fn}^r(S)^{1-\sigma} \right]^{1/(1-\sigma)}. \quad (2.4)$$

This is the cost of the segment's CES service composite under the maintained quality weights. It can rise because suppliers charge more, because a network loses accessibility, or because a previously available service becomes ineligible.⁹

We adopt the share-dependent markup schedule of Atkeson and Burstein (2008), also used by Edmond et al. (2015), as a conduct specification. Atkeson and Burstein derive this schedule under Cournot competition with nested-CES demand. We retain its share-markup relationship while holding expenditure in each customer segment fixed. Down-

⁸The demand system abstracts from customer-specific switching histories and explicit fees for switching or using multiple clouds. These differ from the geographic service costs studied here. Competition and Markets Authority (2025, Chapter 6) documents technical barriers and egress fees affecting switching and multi-cloud use.

⁹The index allows service availability and accessibility to vary, so it need not move with posted prices for continuing services. Feenstra (1994) studies the related role of changing varieties in CES price measurement. Security benefits and network choices by unmodeled suppliers are outside the comparison; these price effects do not measure the full welfare consequences of regulation.

stream payoffs follow this pricing system:

$$p_{fn} = \mu(s_{fn})c_{fn}, \quad \mu(s) = \frac{\mathcal{E}(s)}{\mathcal{E}(s) - 1}, \quad \frac{1}{\mathcal{E}(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad \sigma > \eta > 1. \quad (2.5)$$

A larger supplier faces a lower perceived elasticity and charges a higher markup, bounded between $\sigma/(\sigma - 1)$ and $\eta/(\eta - 1)$. With segment expenditure held fixed, η governs the share–markup relationship rather than an expenditure response. The CES comparison in Appendix C.3 instead holds markups constant at $\sigma/(\sigma - 1)$.

Operating profit is

$$\begin{aligned} VP_{fn} &= \frac{R_n s_{fn}}{\mathcal{E}(s_{fn})} \\ &= R_n \left[\underbrace{\frac{s_{fn}}{\sigma}}_{\text{constant-markup term}} + \underbrace{\left(\frac{1}{\eta} - \frac{1}{\sigma} \right) s_{fn}^2}_{\text{variable-markup term}} \right], \quad \Pi_f = \sum_n VP_{fn}. \end{aligned} \quad (2.6)$$

The positive share-squared term reflects the additional profit from the higher markup of a larger supplier. The terms decompose profit at a given share; changing from AB to CES pricing also changes equilibrium shares. A location changes delivered costs and accessibility, but its return also depends on the share and markup obtained after every supplier reprices. Even when two locations are technological substitutes, these market-share effects can alter the incentive to include both in a portfolio.

Proposition 1 (Pricing uniqueness). *For $\sigma > \eta > 1$ and a fixed portfolio profile, the pricing system (2.3)–(2.5) has a unique solution. In markets with at least two active firms, the scalar-response update of log price-index contributions is a global sup-norm contraction. Appendix A.6 defines this update and gives its contraction modulus. Equilibrium prices vary smoothly with strictly positive marginal costs on each fixed active set.*

Adding a fixed residual contribution preserves the bound because it reduces the weight of responsive strategic rivals in each supplier’s competitor aggregate. Unique prices make each portfolio profile’s operating payoff well defined. They do not establish uniqueness of the portfolio equilibrium, which requires a separate bound on portfolio responses.

2.3 Private location costs

A core portfolio incurs the additive net cost

$$C_f(S_f, Z_f) = \ell_f^{\text{ref}} \sum_{i \in S_f} \left[\underbrace{\alpha_f + \delta M_i + \kappa D_{fi}}_{\text{systematic net cost}} + \underbrace{\sigma_\varepsilon Z_{fi}}_{\text{private shock}} \right], \quad Z_{fi} \stackrel{\text{iid}}{\sim} N(0, 1). \quad (2.7)$$

The firm-specific intercept α_f governs worldwide portfolio size: a higher common per-core charge makes larger portfolios less attractive. The intercept may be positive or negative and is interpreted as a reduced-form net cost of core presence, not an estimate of cash construction expenditure. An intercept below zero represents a net benefit before distance and missing-data penalties. It can absorb returns omitted from modeled IaaS profits, such as other services, but the calibration does not identify the source of that benefit. Any such absorbed return is held fixed in the counterfactuals. Negative private-cost realizations are also retained.

Here D_{fi} is distance from the firm's home country. For OVHcloud, whose home market is the model EU, it is distance from the nearest model-EU country and hence zero within that region. The indicator M_i records a missing original construction-cost observation. Costs are expressed in operating-profit units using a fixed provider-specific loading $\ell_f^{\text{ref}} = G_f^{\text{ref}} / K_f^{\text{ref}}$, the ratio of reference operating profit to a reference core count.¹⁰ Section 4 states the maintained values and the calibration of α_f .

Portfolio size is endogenous.¹¹ A policy changes the eligibility or delivered cost of service paths; the resulting portfolio response can change both the number and location of core-hosting countries. The counterfactuals compare static equilibria under different policy regimes, without tracing the transition between them.

¹⁰The reference values are fixed inputs, not the profit and core count of the portfolio being evaluated. Holding the other coefficients fixed, changing ℓ_f^{ref} changes both deterministic costs and private-cost dispersion, and can therefore change location choices. It is a maintained scale assumption, not merely a relabeling of units. Appendix D reports the reference values.

¹¹A lower expected core count denotes a different network configuration, not a prediction of immediate facility closure. Installed local-service topology and operating arrangements remain fixed. Construction delays and transitions between network configurations would require a dynamic framework, such as Ericson and Pakes (1995).

3. STRATEGIC LOCATION CHOICE AND EQUILIBRIUM

Section 2 determines operating profits for any configuration of provider networks. We now study how firms choose those networks when each location's profitability depends on the rest of the portfolio and on competitors' choices. After defining the portfolio game, we show how a location's net return can change sign as its network expands and construct profit bounds for best responses over all feasible portfolios. The general game and response conditions also apply to other finite downstream payoffs; the cloud structure supplies the more specific optimization bounds. The final subsection gives an additional condition for equilibrium uniqueness.

3.1 Portfolio game and equilibrium

There are F firms. Firm f has a finite set $\mathcal{A}_f$ of portfolio elements and a nonempty feasible family $\mathcal{X}_f \subseteq 2^{\mathcal{A}_f}$. For every pure portfolio profile $S = (S_1, \dots, S_F)$, let $p^+(S)$ denote a fixed selection of the downstream outcome and $\Pi_f(S; p^+(S))$ the associated operating profit. The selection rule allows multiple downstream equilibria in the general game; the cloud pricing system in Section 2.2 determines this outcome uniquely. The selected payoff, net of deterministic costs of portfolio elements, is

$$r_f(S) = \Pi_f(S; p^+(S)) - \sum_{j \in S_f} C_{fj}^0. \quad (3.1)$$

In the cloud application, each portfolio element is a core-location choice; the service connection between a local facility and its parent core is a different object. Here C_{fj}^0 is the deterministic net cost of element j . It is a separate primitive from the positive private-shock loading ℓ_{fj} . The cloud specification is obtained by setting

$$C_{fj}^0 = \ell_f^{\text{ref}}(\alpha_f + \delta M_j + \kappa D_{fj}), \quad \ell_{fj} = \ell_f^{\text{ref}}, \quad \varepsilon_{fj} = Z_{fj}.$$

These substitutions recover (2.7).

Firm f observes the vector of private cost shocks $\varepsilon_f = (\varepsilon_{fj})_{j \in \mathcal{A}_f}$. Its payoff is

$$u_f(S; \varepsilon_f) = r_f(S) - \sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj}, \quad \ell_{fj} > 0. \quad (3.2)$$

Throughout, $\sigma_\varepsilon > 0$. The loading ℓ_{fj} converts the standardized private-cost shock into payoff units.

Assumption 1 (Private cost shocks). Private types are independent across firms, and each firm's type vector has an absolutely continuous joint distribution. Within a firm, the coordinates may be dependent. For every j , the conditional density of ε_{fj} given $\varepsilon_{f,-j}$ is bounded by $\bar{g}_{fj} < \infty$ almost surely. Independent normal or logistic cost shocks satisfy this condition.

Private costs represent residual information the firm knows better than its rivals, such as negotiated site terms or implementation requirements. Common wages and regulation can enter public costs and operating profits; independence across firms applies to the remaining private components. Within-firm location costs may be correlated, and a shared location shock already correlates the costs of overlapping portfolios. Private profitability information is familiar in entry games (Seim, 2006); Sabal (2026) combines it with additive product costs in a portfolio game.¹²

Additivity excludes private cost economies of scope, but allows locations to complement or substitute for one another through operating profits. Absolute continuity makes ties negligible and supports continuity of choice probabilities. The conditional density cap limits the mass near a location's inclusion cutoff, given other costs. It is needed for response bounds, not for optimizing a fixed type's portfolio.

Let $Q_f \in \Delta(\mathcal{X}_f)$ be firm f 's portfolio distribution before its private costs are observed, and write $Q_{-f} = \bigotimes_{g \neq f} Q_g$. In equilibrium, each type chooses an optimal portfolio, and the type distribution induces Q_f . Firm f 's interim value of portfolio S_f , averaging over rivals' choices, is

$$V_f(S_f; Q_{-f}) = \sum_{S_{-f}} r_f(S_f, S_{-f}) Q_{-f}(S_{-f}). \quad (3.3)$$

¹²Sabal uses positive lognormal fixed costs; the cloud application uses signed net costs with Gaussian private components. The common features are private information and costs attached to portfolio elements, not their distributions.

Independence across firms gives the product Q_{-f} .

The type-specific global best response and the distribution it induces are

$$\text{BR}_f(\varepsilon_f; Q_{-f}) \in \arg \max_{S_f \in \mathcal{X}_f} \left\{ \underbrace{V_f(S_f; Q_{-f})}_{\text{payoff net of deterministic costs}} - \underbrace{\sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj}}_{\text{realized private costs}} \right\}, \quad (3.4)$$

$$\mathcal{B}_f(Q_{-f})(S_f) = \Pr\{\text{BR}_f(\varepsilon_f; Q_{-f}) = S_f\}. \quad (3.5)$$

Definition 1 (Portfolio Bayesian Nash equilibrium). A portfolio Bayesian Nash equilibrium is a profile $Q^* = (Q_1^*, \dots, Q_F^*)$ satisfying

$$Q_f^* = \mathcal{B}_f(Q_{-f}^*), \quad f = 1, \dots, F.$$

Each type chooses a globally optimal portfolio in $\mathcal{X}_f$, with ties resolved by a fixed rule.

Proposition 2 (Existence). *Suppose the continuation selection in (3.1) is fixed independently of entry-cost types and beliefs, all selected payoffs are finite, types are independent across firms, and each within-firm type vector has an absolutely continuous joint distribution. Then a portfolio BNE exists.*

Appendix A.1 gives the proof. Absolute continuity makes ties negligible and the distributional response map continuous, so Brouwer's theorem applies.

3.2 Why location returns can change sign

An additional location can improve a firm's ability to serve several markets, but its return depends on the network already in place. At low supply, expanding that network can increase the return to a further location. At high supply, diminishing gains in market share can reverse this relation. Under variable markups, competitors' repricing changes the comparison as well. The following one-market calculation isolates these forces before introducing private costs.

Let $x > 0$ summarize a firm's supply in one market and let its pre-markup competitiveness be $\xi_i = \psi_i c_i^{1-\sigma} = \omega_i x^{\nu_s}$, where $\omega_i > 0$ collects factors held fixed as supply changes. Rival competitiveness, residual supply, and expenditure are fixed, but all suppliers reprice

Table 1: Local profit curvature differs at the same market share

Other strategic suppliers	Residual share	Local curvature
One rival	0	Concave
Seven equal rivals	0	Convex
One rival	0.20	Concave
Seven equal rivals	0.20	Convex

Notes: Constructed AB markets with $(\sigma, \eta, \theta) = (5, 2, 3)$ and own share $s_i = 0.08$. Named competitors divide the share left after own and residual supply. Each row chooses rival primitives to support these shares, then holds them fixed when calculating the own-supply derivative. The rows compare different economies, not a merger experiment. Appendix A.7 reports the corresponding curvature coefficients.

as x changes. In the core-only benchmark without an accessibility margin, $v_S = (\sigma - 1)/\theta$ under (2.1).

The constant-markup case gives the simplest intuition. At $\eta = \sigma$, the sign of the second derivative of operating profit $\Pi_i(x)$ is the sign of

$$\underbrace{(v_S - 1)}_{\text{competitiveness gain at low supply}} - \underbrace{2v_S s_i}_{\text{diminishing scope to gain share}}.$$

For $v_S > 1$, profit is convex at low shares and concave at high shares. An addition can therefore become more valuable as a small network grows, yet less valuable once the provider serves many customers. In this core-only example, low-share convexity requires $(\sigma - 1)/\theta > 1$. Fixed workload composition permits this case: expected unit cost is finite for every $\theta > 0$. If workloads were instead aggregated by CES with elasticity equal to the provider-demand elasticity σ , finiteness would require $\theta > \sigma - 1$ and rule out this particular channel. The internal and external elasticities need not be equal; Appendix D.1 states the general condition.

Under AB pricing, the same comparison also depends on competitors' sizes. The own-share response $D_i = \partial s_i / \partial \log \xi_i$ and its elasticity $E_i = \partial \log D_i / \partial \log \xi_i$ incorporate their repricing. Appendix A.7 derives these responses and the curvature expression (A.19). Unlike the CES expression above, its sign depends on the entire distribution of competing shares.

Table 1 holds own share fixed at eight percent. With one competitor, profit is locally concave in supply; with seven equal competitors, it is locally convex. Thus own share alone

does not determine whether small location additions complement or substitute for one another. For finite additions $d_j, d_k > 0$, the relevant discrete interaction is $\int_0^{d_j} \int_0^{d_k} \Pi_i''(x + u + v) dv du$; its sign depends on the whole interval, not just the initial curvature. Across markets, these interactions are added with the corresponding expenditure weights.

Variable markups add dependence on rival composition; they are not necessary for every failure of ordering. The following result goes beyond local curvature to a strict reversal in the net return to a discrete location.

Proposition 3 (Sign reversals in the return to a location). *Fix any finite collection of positive rival scores and any residual contribution $b \geq 0$ that together provide competing supply. If $\xi_i = \omega_i x^{v_S}$ with $v_S > 1$, there exist a positive base supply, three positive location contributions, and a positive additive cost for location j such that, writing $d_T = \sum_{h \in T} d_h$ and $\Delta_j(T) = \Pi_i(x_0 + d_T + d_j) - \Pi_i(x_0 + d_T) - c_j$, its net marginal returns satisfy*

$$\Delta_j(\emptyset) < 0, \quad \Delta_j(\{k\}) > 0, \quad \Delta_j(\{k, h\}) < 0.$$

Every own subset is feasible, and prices are solved at every portfolio. The conclusion also holds in the constant-markup limit.

The construction uses increasing, bounded profits whose slope vanishes near zero supply when $v_S > 1$. An addition can then be unprofitable near zero, profitable at an intermediate network, and unprofitable again near market saturation. Appendix A.7 gives the proof. The proposition establishes a possible failure for arbitrary fixed competing supply, not its frequency among a given set of locations.

This strict $-, +, -$ pattern matters more than nonmonotone curvature alone. The positive marginal return is lost both when the network contracts and when it expands. It therefore violates both directions of the single-crossing condition used by Arkolakis et al. (2026); rising and then falling marginal returns also rule out both submodularity and supermodularity. Consequently, the portfolio-order restrictions supporting the bounds in Sabal (2026) are not guaranteed by this pricing environment. The next subsection bounds the profit of complete portfolios directly, without these restrictions.¹³

¹³A country-specific inclusion cutoff does not avoid this optimization: it compares the best complete network containing the country with the best network excluding it. Both can change with other location costs. The conditional cutoff is stated in Appendix A.1 and used in the probability-flow proof.

3.3 Completion bounds for portfolio best responses

Suppose a feasible network earns net profit 100. Elsewhere in the search, 30 countries remain undecided after other location decisions have been fixed, giving 2^{30} possible combinations. If a calculation favorable to every remaining network bounds its profit by 92, the whole group can be discarded without pricing its members. A bound of 120 leaves the comparison unresolved. Fixing one country's entry decision then splits the group into two smaller groups, whose bounds can be tighter.

This is the logic of *branch and bound* (Land and Doig, 1960). A good feasible network gives a lower bound on the best attainable profit; a *completion bound* limits what any unexamined completion can earn. Computation is saved when these two comparisons eliminate whole groups. The contribution here is the construction of profit bounds using the service and cost structure of the portfolio problem. Proposition 3 explains why a location's marginal return alone cannot provide the required elimination rule. The bounds below remain valid when that return changes sign as the network expands.¹⁴

Ordering-based elimination instead uses what a marginal return says about other networks. With complementary locations, for example, an addition that is unprofitable in the largest available network is also unprofitable in smaller networks. The sign reversals above invalidate that inference. The search evaluates feasible portfolios using the original payoffs. Its upper bounds may relax constraints and interactions, provided they bound the profit of every feasible completion.

For a type ε_f , denote the objective in (3.4) by $\tilde{V}_f(T_f; \varepsilon_f, Q_{-f})$. A partial-portfolio subproblem (S_f, U_f) records portfolio elements already selected in S_f and those still undecided in U_f . Its feasible completions form the set $\mathcal{X}_f(S_f, U_f)$. This subproblem is a node of the search tree, representing a group of complete portfolios. A completion upper bound is any value $B_f(S_f, U_f)$ that dominates the payoff of every such completion:

$$\mathcal{X}_f(S_f, U_f) = \{T_f \in \mathcal{X}_f : S_f \subseteq T_f \subseteq S_f \cup U_f\}, \quad B_f(S_f, U_f) \geq \max_{T_f \in \mathcal{X}_f(S_f, U_f)} \tilde{V}_f(T_f; \varepsilon_f, Q_{-f}). \quad (3.6)$$

¹⁴Branch and bound is a general search procedure. Its worst-case search can remain exponential in the number of locations; the computational comparisons below measure the gains from our bounds on the reported problems.

If $B_f(S_f, U_f)$ does not exceed the best payoff already found, no portfolio represented by that subproblem can improve on the best choice already found.

An optimistic network with the right number of locations. The first bound retains one feature of the common portfolio: its number of additional elements. If the firm adds k locations, give it the k largest contributions to each market statistic and charge only the k lowest location costs. Finding these contributions requires sorting lists of locations, not evaluating every k -location network. The selected countries may differ across statistics and costs, making the calculation deliberately optimistic. Comparing the resulting bounds over all feasible k allows the total number of locations to be chosen endogenously. Write the objective (3.4) equivalently as $\sum_{\varsigma} q_{\varsigma} \pi_{f\varsigma}(T_f) - \sum_{j \in T_f} C_{fj}$, separating operating profit from total costs of portfolio elements: ς indexes a finite rival-portfolio state, $q_{\varsigma} \geq 0$ is the probability the rival portfolio distribution assigns to state ς (a point mass on the realized rival profile in a complete-information benchmark), and $C_{fj} = C_{fj}^0 + \sigma_{\varepsilon} \ell_{fj} \varepsilon_{fj}$ combines deterministic and realized private costs. Suppose the continuation payoff in state ς is bounded by a coordinatewise increasing function of nonnegative additive statistics, indexed by $m = 1, \dots, L$:

$$z_{\varsigma m}(T_f) = z_{\varsigma m}^0 + \sum_{j \in T_f} a_{\varsigma m j}, \quad a_{\varsigma m j} \geq 0, \quad \pi_{f\varsigma}(T_f) \leq \bar{\pi}_{f\varsigma}(z_{\varsigma 1}(T_f), \dots, z_{\varsigma L}(T_f)).$$

Monotonicity is required of the bounding function, not of the actual marginal profits from adding a location. The bound may rise with available supply even when the additional profit from a particular location rises and falls as the network expands.

Let $a_{\varsigma m, (h)}^{\downarrow}(U_f)$ be the h th largest contribution to statistic (ς, m) among undecided elements, and let $C_{f, (h)}^{\uparrow}(U_f)$ be the h th smallest element cost. Define

$$\begin{aligned} \bar{z}_{\varsigma m}(S_f, U_f, k) &= \underbrace{z_{\varsigma m}(S_f)}_{\text{base and selected locations}} + \underbrace{\sum_{h=1}^k a_{\varsigma m, (h)}^{\downarrow}(U_f)}_{\text{best } k \text{ contributions}}, \\ \underline{C}_f(S_f, U_f, k) &= \underbrace{\sum_{j \in S_f} C_{fj}}_{\text{selected locations}} + \underbrace{\sum_{h=1}^k C_{f, (h)}^{\uparrow}(U_f)}_{\text{cheapest } k \text{ additions}}. \end{aligned}$$

Proposition 4 (Cardinality-conditioned completion bound). *Under the conditions above,*

$$B_f^{\text{card}}(S_f, U_f) = \max_{0 \leq k \leq |U_f|} \left\{ \underbrace{\sum_{\zeta} q_{\zeta} \bar{\pi}_{f\zeta}(\bar{z}_{\zeta}(S_f, U_f, k))}_{\text{operating-profit bound with } k \text{ additions}} - \underbrace{\underline{C}_f(S_f, U_f, k)}_{\text{total cost lower bound}} \right\} \quad (3.7)$$

is an upper bound on every feasible completion represented by the subproblem. The result applies to any finite feasible portfolio family; including cardinalities with no feasible completion only weakens the bound.

A common number k of additions prevents crediting all remaining locations while paying for only a few. Maximizing over k permits endogenous portfolio size, including negative realized costs of portfolio elements. Appendix B gives the proof and the CES and AB envelopes. Local service can require a parent core, and core and local supply in one country do not add. Supply is therefore not additive in the chosen core set. The bound uses an additive function that dominates this service mapping at every portfolio.

Making benefits and costs come from the same locations. Figure 1 illustrates the source of slack with one additional core and two customer markets. Separate maxima credit Germany’s contribution in Europe and Singapore’s contribution in Asia while charging Poland’s cost. Their sum, 17, is an upper bound, but no single core can deliver it. Requiring the same country to provide both benefits and pay its own cost lowers the bound to 7 in this linear example. If a network elsewhere in the search already earns 8, the entire group is excluded.

The joint-affine refinement ranks countries by a common affine bound on their total benefit net of cost, preventing one country’s benefit from being paired with another country’s cost. Proposition B.1 in Appendix B.2 gives the construction and proves that its bound is no larger than the cardinality bound built from the same component envelopes. These bounds combine standard branch-and-bound and top- k tools (Feldman and Topaloglu, 2015; Leitner et al., 2024; Zhang et al., 2025) with restrictions linking market benefits and location costs across rival states.

A tractable relaxation of the common network. In the application, market profit is non-linear and local service may depend on a parent core. We retain these connections through

One additional core must serve both customer markets

Core location	Europe benefit	Asia benefit	Cost	Net profit
Germany	9	1	6	4
Singapore	2	9	5	6
Poland	4	4	1	7

Separate market choices	One common location choice
Germany for Europe; Singapore for Asia; Poland for cost. $9 + 9 - 1 = 17$ An optimistic, infeasible combination.	Each country's benefits are combined with its own cost. $\max\{4, 6, 7\} = 7$ A bound below a known profit of 8.

Figure 1: Why common location choices tighten a profit bound

Notes: Hypothetical payoffs in arbitrary units. In this illustrative subproblem, one core must be chosen and operating profit is the sum of the two market benefits. A feasible portfolio outside this group is assumed to earn 8. The separate-market bound cannot exclude the group, whereas the common-choice bound can. These values are not calibration inputs or application results. With nonlinear profits and fractional choices, the common-choice bound need not equal the best feasible profit.

common location and service variables, while making the remaining optimization easier in two ways: fractional core choices are allowed, and the repriced profit curve in each market is replaced by a concave function that lies above it. Every actual network remains feasible in this enlarged problem and is credited with at least its actual profit. Its optimized value is therefore an upper bound. Fractional facilities are used only for this calculation; the firm still chooses a discrete portfolio.

The two statistics are already present in the cloud model. Suppressing firm and policy indices, equations (2.1)–(2.2) give

$$\Sigma_n = \Sigma_{fn}, \quad W_n = 1 + \sum_i o_{fin} e^{\lambda_A q_{in}}, \quad \xi_n = \psi_{fn} c_{fn}^{1-\sigma} = \zeta_n \Sigma_n^{\nu_S} W_n^{\nu_A},$$

where $\nu_S = (\sigma - 1)/\theta$, $\nu_A = \chi/\lambda_A$, and $\zeta_n = a_f^{\sigma-1} \exp(d_{fn} - \chi A_n^{\text{frontier}})$. Thus the same service choices determine both effective supply and access. Let x_j indicate a core choice, let v collect service-coverage variables, and set $\vartheta = \nu_S/(\nu_S + \nu_A)$. For $\nu_S, \nu_A > 0$, define

$$z_n = \Sigma_n^{\vartheta} W_n^{1-\vartheta}, \quad \xi_n = \zeta_n z_n^{\nu_S + \nu_A}. \quad (3.8)$$

This is an exact change of variables. The combined statistic is concave in supply and accessibility even when repriced profit is not. At fixed rivals' distribution, let $H_n(z)$ be expected operating profit in segment n at own score $\zeta_n z^{v_S + v_A}$, repricing all suppliers in every rival state before averaging.

Proposition 5 (Completion bound with common location choices). *For a node and completion size k , let $\mathcal{R}_f(S_f, U_f, k)$ be a compact polytope containing a representation (x, v) of every feasible completion, with affine supply $\Sigma_n \geq 0$ and accessibility $W_n > 0$ throughout it. For each completion T , its representation has $x = \mathbf{1}_T$, and these statistics equal those generated by its original core–local service mapping. Let $\hat{H}_n$ be a continuous, nondecreasing concave upper bound on H_n on a compact interval $I_n \subseteq [0, \infty)$ containing every feasible z_n . Then*

$$\begin{aligned}
B_{f,k}^{\text{geo}} = \max_{x,v,z,h} \quad & \underbrace{\sum_n h_n}_{\text{relaxed operating profit}} - \underbrace{\sum_{j \in \mathcal{A}_f} C_{fj} x_j}_{\text{cost of common choices}} \\
\text{subject to} \quad & (x, v) \in \mathcal{R}_f(S_f, U_f, k), \\
& z_n \in I_n, \quad 0 \leq z_n \leq \Sigma_n(x, v)^{\delta} W_n(x, v)^{1-\delta}, \\
& h_n \leq \hat{H}_n(z_n) \quad \text{for every } n
\end{aligned} \tag{3.9}$$

is a convex optimization problem whose value bounds every such completion's profit. Maximizing over feasible k permits endogenous portfolio size.

The same core and service choices now determine every market's supply, accessibility, and location costs. Fractional choices remain possible, but the bound cannot choose a different network separately for each market. Appendix B.4 proves the result, constructs the service constraints, and gives a linear-programming relaxation. Combining any valid bound from this construction with the earlier bounds by taking their minimum cannot weaken the upper bound; the new bound alone need not dominate them. The market-profit curves can be reused across private-cost types at fixed rivals and policy, although their construction and the optimization both take time. Figure 2 measures whether this additional preparation pays off. Using the calibration in Section 4, with 20 optional countries in the eight-provider model, the common-location bounds find the same optimal profits as exhaustive enumeration a median 235 times faster, including preparation, and are 10.6 times faster than the cardinality and joint-affine bounds. Preparation can out-

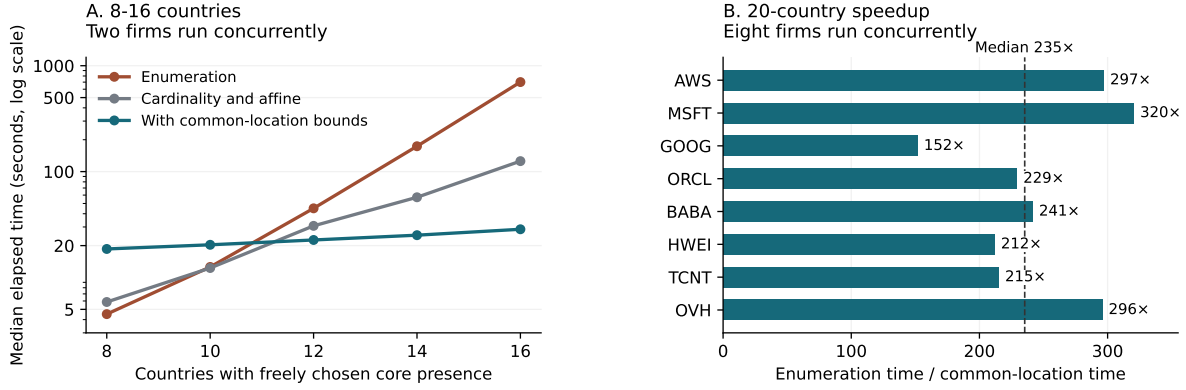


Figure 2: Portfolio search compared with exhaustive enumeration

Notes: The sample comprises all eight providers, one prespecified nested sequence of optional countries, and two private-cost vectors per firm. Other core choices are fixed to the observed network. Times include setup and the solution of both types; each method can reuse its own operating-profit evaluations. All searches finish, and optimal profits agree across methods. Panel A reports median elapsed times with two firms running concurrently. Panel B reports enumeration/common-location-bound time ratios with eight firms running concurrently; each method uses one computation thread. The different concurrency settings are kept in separate panels. Each comparison uses a single timing run.

weigh the savings in small problems. These comparisons optimize against given rivals, holding other locations fixed; they do not measure the time to solve an unrestricted 42-country problem or an entire equilibrium. Appendix C.2 gives the protocol and timings.

From optimal networks to a portfolio equilibrium. Fix the competing firms' portfolio distributions. For each own-cost type, the search compares complete networks using profits averaged over those rival distributions. Aggregating the optimal choices over private costs gives $\mathcal{B}_f(Q_{-f})$ in (3.5). A country may be chosen more often without becoming independent of the firm's other locations; the object passed to competing firms is the distribution of whole networks.

We iterate distributions over a finite menu of candidate networks. At the resulting profile, the global search looks for profitable networks outside the menu. Any discovered network is added to every regime's menu within the same application before the distributions are recomputed. At the final profile, the relevant tests are mutual consistency of the distributions and bounds on gains from deviating to any feasible portfolio. An optimum against an earlier rival distribution does not establish optimality against the final one.

The bounds above make the inner portfolio search globally valid; they do not by themselves ensure convergence of the outer distribution updates. The next subsection gives

an additional sufficient condition for that convergence and for uniqueness. Appendix B.5 distinguishes global-search error from integration and equilibrium approximation errors.

3.4 An additional condition for equilibrium uniqueness

Private-cost dispersion can limit strategic feedback (Mason and Valentinyi, 2007; Melo, 2022): a payoff change alters choices only for types sufficiently close to a boundary between optimal portfolios. Overlapping portfolios share location costs, so these boundaries must be considered jointly.

For one optional location with deterministic net benefit a , inclusion has probability $F_\varepsilon(a/(\sigma_\varepsilon \ell))$. If the shock density is bounded by $\bar{g}$, changing the benefit by Δa changes inclusion probability by at most $\bar{g}|\Delta a|/(\sigma_\varepsilon \ell)$. The portfolio counterpart assigns changes in relative payoffs to the locations on which competing portfolios differ. Appendix A.2 constructs the resulting response-bound matrix $K^\sharp$ using a pair-cover argument: its entry $K_{fg}^\sharp/\sigma_\varepsilon$ bounds the change in firm f 's response per unit change in rival g 's distribution, measured in total variation. It covers all feasible portfolios and rival distributions. Independent normal costs permit a further refinement $K^N \leq K^\sharp$ by absorbing part of a payoff change into a shift of the cost vector; K^N can replace $K^\sharp$ in the condition below.

Theorem 1 (Uniqueness under sufficiently dispersed private costs). *Under the assumptions in Section 3.1, if*

$$\underbrace{\sigma_\varepsilon}_{\text{private-cost dispersion}} > \underbrace{\rho(K^\sharp)}_{\text{bound on strategic feedback}}, \quad (3.10)$$

the portfolio BNE is unique, and simultaneous distributional best-response iteration converges to it from every initial distribution.

The spectral radius measures feedback around cycles. With two firms, let $M_{fg} = K_{fg}^\sharp/\sigma_\varepsilon$. Distances between two candidate equilibria would satisfy $d_1 \leq M_{12}d_2$ and $d_2 \leq M_{21}d_1$. If $M_{12}M_{21} < 1$, both distances must be zero. The matrix condition extends this logic to more firms. Appendix A.5 proves the result and gives the corresponding error bound for an approximate equilibrium. A small AB game in Appendix C.1 satisfies the refined uniqueness condition even though a location's net return changes sign in both directions along nested portfolios. The failure of an ordering condition therefore need not imply multiple equilibria.

Total variation counts changes between portfolios even when they have similar market effects. An alternative condition in Appendix G.1 retains market effects and signed responses, with an explicit allowance for omitted interactions; it establishes uniqueness without guaranteeing convergence of response iteration. Neither condition is verified for the calibrated applications. In particular, Appendix C.4 shows that the global normal-cost condition fails at the baseline parameters.

4. QUANTIFYING MULTINATIONAL CLOUD NETWORKS

We calibrate provider-specific location-cost intercepts to worldwide core counts and relative competitiveness and residual supply to global shares. These targets discipline worldwide scale; geographic allocation and policy responses also depend on the assigned spatial costs, service connections, and private-cost dispersion. Regional composition and hosting-country characteristics assess those implications.

4.1 Data and assigned inputs

Our 2026 eight-provider sample covers Amazon Web Services, Microsoft Azure, Google Cloud, Oracle Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud, and OVHcloud. It records 169 core-country presences in a 42-country universe of 336 provider-country choices. As in Section 2.1, multiple core regions of the same provider in one country count as a single presence. Every country is available as a core location in the EU specification, including unobserved and unannounced locations. The China application adds explicit operating-mode restrictions.

The model EU contains Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, the Netherlands, Poland, Spain, and Sweden.

Local-service records distinguish core presence from service availability. The screened sample contains 60 local records: 34 have documented parent regions and 26 have unconfirmed parents. Among the 33 documented AWS connections, nine cross a national border and 24 are within-country.¹⁵

¹⁵The remaining confirmed parent is Microsoft’s Los Angeles Extended Zone. The records with unconfirmed parents comprise 25 OVHcloud records and one Alibaba record. These are documentation counts, not capacity, traffic, or industry-wide frequencies. Appendix E.1 records the sources and coverage.

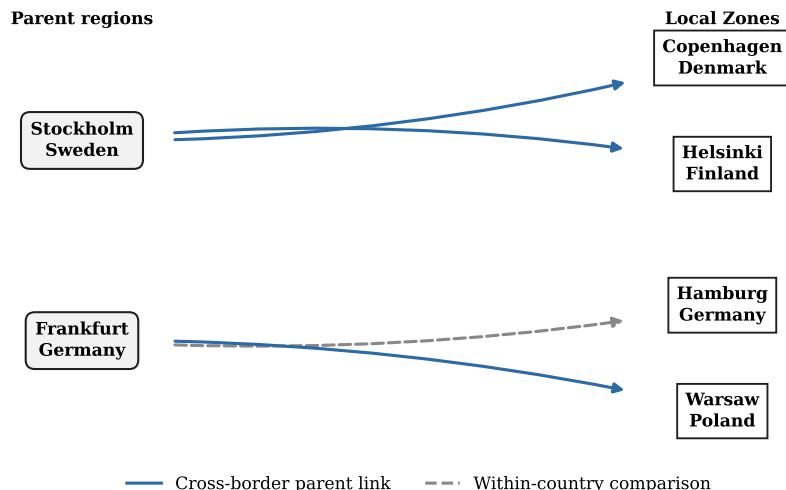


Figure 3: A core region can support local cloud services in another country

Notes: Solid blue lines connect AWS Local Zones to parent regions abroad; the dashed gray line stays within Germany. Lines show parent-region relationships, not necessarily data routes. Sources: AWS Local Zone roster and architecture documentation, archived September 11, 2026.

Figure 3 illustrates the distinction: AWS’s Copenhagen Local Zone has Stockholm as its parent region, whereas Hamburg–Frankfurt provides a within-country comparison. We therefore record core locations and the countries with local service separately.

Worldwide network size and regional presence can have different orderings. OVHcloud has cores in ten sample countries and Alibaba in 17, yet OVHcloud has four core presences in the 12 model-EU countries compared with Alibaba’s two.

Expenditure is proportional to $(\text{GDP} \times \text{internet-use rate})^{0.7}$, normalized to one in the United States. Appendix D.2 gives its formula and source conventions. Splitting each model-EU country into regulated and private demand gives 54 customer segments. The regulated fraction $\omega = 0.18$ sets the assumed exposure of EU expenditure to the procurement rule. Both segments share supply rules in the baseline.¹⁶

The operating input-cost index is $w_i = 0.75 + 0.20(p_i^E/p_{US}^E) + 0.05(L_i/L_{US})$. Electricity prices use large industrial-user tariffs, including Eurostat band IF where available, and

¹⁶The exposure share assigns 18 percent of total modeled EU cloud expenditure to the regulated segment as a scenario assumption, rather than an estimate from procurement spending. The Commission’s figures for the scope of SecNumCloud concern shares of public-sector use cases, not shares of total cloud expenditure (European Commission, 2026b, Part 1, p. 43). They therefore do not directly calibrate this parameter or establish its order of magnitude.

documented fallbacks. Labor costs use a human-capital-adjusted compensation proxy. Here p_i^E is the electricity price and L_i the labor-cost index.

Delivered-service costs use a three-level geographic rule:

$$\tau_{in} = \begin{cases} 1, & i = n, \\ 1.15, & i \neq n, \text{ same continent or within 5,000 km,} \\ 1.50, & \text{otherwise.} \end{cases} \quad (4.1)$$

At $\theta = 3$, these foreign delivery wedges give an origin approximately 66 or 30 percent of an otherwise identical domestic origin's contribution to the cost aggregator, since $1.15^{-3} \simeq 0.66$ and $1.50^{-3} \simeq 0.30$. The delivery-distance rule uses the input file's city reference coordinates, rather than individual facilities. Thus all foreign EU-to-EU origins have the same baseline cost wedge.

Measured latency instead disciplines which foreign EU origins offer better accessibility to a given EU destination. Shorter measured round-trip times receive higher accessibility scores, while preserving the destination's existing distribution of scores. This uses RIPE anchor-to-anchor measurements to rank origins, not to estimate a willingness to pay for lower latency or a provider's private-network performance. Domestic scores and pairs outside the EU block retain the geographic specification. Appendix D.2 gives the ranking procedure and sample conventions. Table 2 links the calibrated and assigned inputs to their economic roles. The elasticities $(\sigma, \eta, \theta) = (5, 2, 3)$ are maintained values. Here σ governs substitution across providers, η the share-markup schedule, and θ the cost reduction from combining supply origins. The fixed-workload production structure in Section 2.1 separates these margins: $\theta > 0$ ensures a finite unit cost, while $\sigma > \eta > 1$ governs pricing. Together with $(\chi, \lambda_A) = (0.4, 1.5)$, these values imply supply and accessibility score exponents $\nu_S = 4/3$ and $\nu_A = 4/15$; their sum, 1.6, is the exponent on the combined score in Section 3.3.

The demand shifter d_{fn} equals $\log 4$ for U.S. providers serving U.S. customers, Chinese providers serving China, and OVHcloud serving the model EU; otherwise it is zero. OVHcloud's preference is regional. At given costs and prices, home preference raises a provider's price-index contribution by a factor of four; equilibrium shares do not increase fourfold. Relative competitiveness is calibrated conditional on this home advantage.

In the net cost of core presence (2.7), distance is measured in 10,000 kilometers. We maintain $\delta = 0.614$, $\kappa = 0.5$, and $\sigma_\varepsilon = 0.5$. The level of the construction-cost index does not enter (2.7); its missing-data indicator does. An additional 10,000 kilometers raises deterministic cost by $0.5\ell_f^{\text{ref}}$ per core, equal to one standard deviation of that provider’s private per-core cost shock. There is no regional cost intercept. The missing-data penalty is inherited from a location-composition fit that held provider core counts fixed; Appendix D.3 gives its role in the present calibration.¹⁷ Fixed provider-specific loadings convert these standardized costs into operating-profit units; Appendix D reports their values and normalization. The loadings remain unchanged when the number of chosen locations changes. Private-cost dispersion is set independently of the uniqueness condition.

Table 2, Panel B, collects the assigned inputs. They are held fixed while worldwide counts and shares discipline the parameters in Panel A. Appendix D records their construction and scope; Section 5 examines residual supply, eligibility, and starting geography.

4.2 Matching worldwide scale and shares

We match expected worldwide core counts using eight intercepts α_f . Seven relative competitiveness parameters, normalized by $a_{\text{AWS}} = 1$, and one residual-supply intensity match eight nonredundant global-share ratios. The residual represents unmodeled suppliers. Its price-index contribution is assumed equal across markets, and this common level is calibrated; its realized share is endogenous. The competitiveness parameters absorb relative supply and demand advantages conditional on the other inputs and are not separately identified physical productivities.

Table 3 lists the targets. Five providers’ share targets come from public 2024 worldwide IaaS data (Gartner, 2025). These shares anchor provider scale in a broader market than the representative workload class. The public data do not identify workload composition or revenues attributable to core and local facilities, so the shares are aggregate calibration targets rather than product-level demand observations. The targets for Oracle, Tencent,

¹⁷Missing construction-cost data need not imply higher location costs. The term δM_i imposes that association; it is not an estimated construction-cost effect. Its economic implications depend on which locations lack data. Belgium is the only model-EU country affected; Appendix D.3 lists all affected countries.

Table 2: Calibrated parameters and assigned inputs

Parameter or input	Value	Economic role and parameter choice
<i>Panel A. Calibrated parameters</i>		
α_f (eight providers)	$(-.119, -.040, .365, .498, .675, .798, .746, 1.268)$	Worldwide core counts discipline each provider's net location-cost level.
a_f/a_{AWS}	$(1, .817, .628, .514, .692, .665, .528, .210)$	Seven relative competitiveness parameters match global-share ratios jointly with residual intensity; $a_{\text{AWS}} = 1$.
Residual intensity	$b_n = b^0 e^{\zeta_b} \simeq 2.045$	Match the residual share with a common price-index contribution across markets.
<i>Panel B. Assigned parameters and inputs</i>		
(σ, η, θ)	$(5, 2, 3)$	Assigned substitution and supply curvature govern pricing and returns to expanding the network.
Missing-data loading	$\delta = .614$	Penalty for missing construction-cost locations; retained from a location-composition fit with core counts fixed.
Home-country distance	$\kappa = .5$	Net location-cost gradient in distance from the provider's home.
σ_ε	.5	Assigned dispersion governs how many types change networks when profits change; counts alone do not identify it.
γ_{loc}	.5	Contribution of local-only service relative to a core.
Accessibility	$\lambda_A = 1.5, \chi = .4$	Accessibility mapping in the price-index contribution.
Home preference	$d_{fn} = \log 4$ at home; 0 elsewhere	Domestic demand advantage; the home market for OVHcloud is the model EU.

Notes: Firm vectors are ordered AWS, Microsoft, Google, Oracle, Alibaba, Huawei, Tencent, and OVHcloud. The parameters are held fixed across policy regimes in both applications. Values are rounded for display. The specification includes no regional cost intercept. The residual level uses $b^0 = 1.77817$ and $\zeta_b = 0.1396$ (rounded).

and OVHcloud are assumed because the public source used here does not report their shares; the residual is the remaining share. These targets are not a complete observed 2026 eight-provider distribution. The infrastructure and share data refer to different years, and detailed country-level revenue shares are not matching targets.

The two matching blocks are coupled through portfolio choice and pricing. We choose cost intercepts to match expected counts, solve the global-share equations for competitiveness and residual intensity, and update portfolio distributions and prices as the parameters change. Worldwide shares average expenditure shares over priced portfolio profiles; pricing an average network would give a different object. The resulting parameters and

Table 3: Worldwide market shares and core-country counts

Provider	World share (%)		World core count	
	Target	Model	Target	Model
AWS	37.7	37.66	28	27.99
Microsoft	23.9	23.85	33	32.97
Google	9.0	8.94	29	28.91
Oracle	4.0	3.98	27	26.92
Alibaba	7.2	7.09	17	16.84
Huawei	4.1	4.05	12	11.85
Tencent	2.0	2.37	13	14.86
OVHcloud	0.1	0.10	10	9.57
Eight providers	88.0	88.03	169	169.92
Residual	12.0	11.97	—	—

Notes: Model values are evaluated at the calibrated parameters and the computed baseline portfolio distribution. Shares average two independently scrambled samples of 2,048 portfolio profiles; expected core counts are calculated directly from the distribution. The maximum share error is 0.37 percentage points and the core-count RMSE is 0.68 countries. Both worldwide counts and share ratios are calibration targets. Appendix F reports numerical accuracy and the remaining search gaps.

portfolio distribution must jointly satisfy the count and share targets and the firms’ best-response conditions. This simulation-based calibration nests count and share matching; it does not treat the 336 entry decisions as independent observations in a likelihood. Calibrated parameters are held fixed across policy regimes.

4.3 Model fit and numerical accuracy

Worldwide scale and shares fit closely (Table 3). The maximum absolute share error is 0.37 percentage points, the core-count RMSE is 0.68 countries, and total expected core presence is 169.92 against 169 observed. Tencent’s worldwide core count remains above target, at 14.86 against 13. Geographic fit is less accurate: the model predicts 52.54 EU presences against 40, a 31.3 percent overprediction. The baseline targets worldwide scale using common spatial-cost coefficients. Regional composition remains a diagnostic rather than being fitted with an EU-specific cost intercept; this choice leaves a substantial geographic mismatch.

These discrepancies matter because the policy experiments change which origins can serve regulated demand. Figure 4 distinguishes core presence from service coverage: the

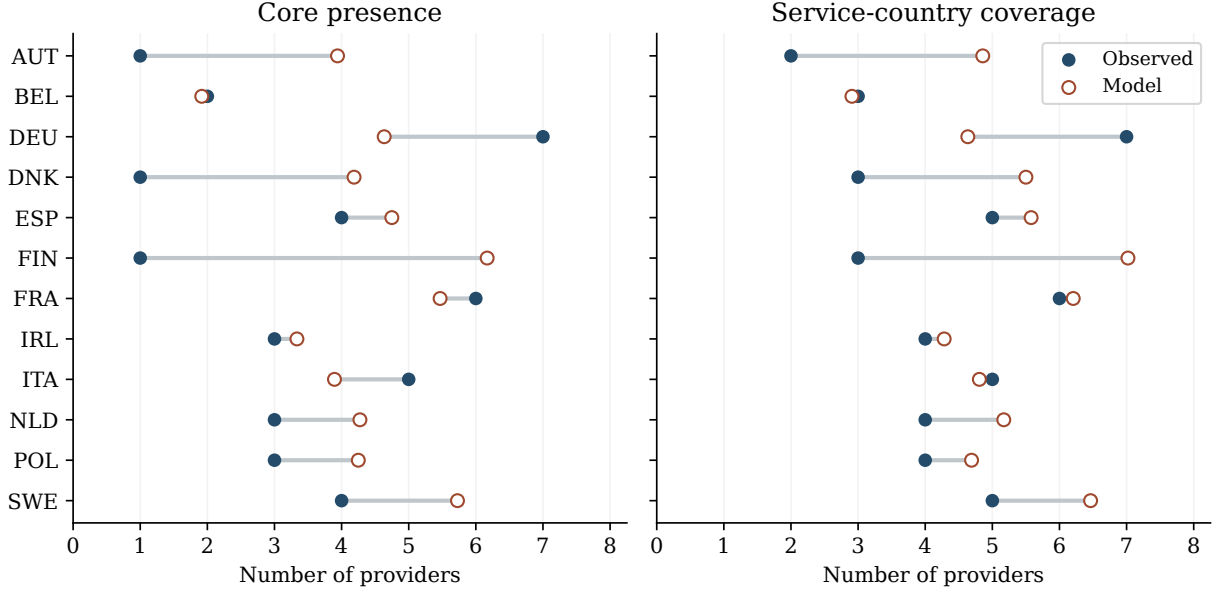


Figure 4: EU geography: core presence and service coverage

Notes: Observed provider counts and expectations under the calibrated baseline portfolio distribution. Both panels use the same country universe and local topology. Model coverage evaluates whether the required parent cores are present for each complete portfolio before averaging. The panels are geographic diagnostics, not additional independent estimation targets.

model underpredicts cores in Germany and overpredicts them in Finland, Austria, and Denmark, while local paths can preserve coverage without a domestic core. Table 6 assesses the fixed-network price effect using observed geography. It changes the entire network and does not isolate the contribution of excess EU presence.

Appendix D.4 reports forty composition diagnostics, which informed specification assessment but are not fitting targets. They show that the model compresses regional specialization: it overpredicts the EU share of core locations for AWS, Oracle, and the Chinese providers and underpredicts it for OVHcloud.

Numerical implementation. The counterfactuals approximate the portfolio equilibrium using the computation described in Section 3.3 and finite candidate sets of complete networks and simulated private costs. Both locations and core counts can change. Joint pricing precedes averaging over profiles, preserving the core-presence events that support local service. Cardinality and joint-affine searches over all feasible portfolios at selected cost types expand the common candidate sets across regimes within each application. The

policy results below use these searches; the additional common-location bound is used in the timing comparison in Figure 2. Policy price effects are $100[\exp(\Delta \log \overline{P}) - 1]$, where the change in expected log price indices uses fixed expenditure weights across customer segments.

We assess candidate-set optimality, distributional consistency, and deviations over the full choice set using, respectively, candidate-set regret, the undamped distributional residual, and the gap between an upper bound on attainable profit and the best feasible profit found by unrestricted search. Rival integration and own-cost sampling are examined separately. Each search bounds remaining improvement for its cost type and rival distribution; selected searches do not establish equilibrium for all continuous types. Higher rival-integration accuracy leaves the reported policy differences close to their original values. The stability of equilibrium policy effects across independent private-cost samples has not yet been established: frozen decision rules evaluated on new types imply five to eleven fewer worldwide core presences. This does not re-solve the equilibria or bound error in paired policy differences. Appendix F reports the corresponding checks and search gaps for these policy profiles. The general uniqueness conditions are not verified for the calibrated portfolio game.

5. INTERNATIONAL NETWORK ADJUSTMENT

The two applications ask how changes in market access reshape multinational portfolios when firms respond to one another. The EU experiment changes which service origins can serve a group of public customers. The China experiment changes the cost of serving mainland customers from offshore facilities. Both allow core-country counts and locations to adjust while retaining the calibrated parameters. The key distinction is between a firm’s preferred network when competitors keep their baseline portfolios and its preferred network when they adjust as well. We separate both from the immediate effect of the restriction at fixed networks. This reveals why similar prices in the directly affected market can conceal different adjustments elsewhere in the global network.¹⁸

¹⁸Appendix Table 18 illustrates the cross-border incidence of both experiments by separating effects on US providers, US customers, and cores located in the United States.

The EU calculation allows cross-border supply within each provider. The China calculation conditions on existing operating modes and partner paths, and its baseline distribution is recomputed under those arrangements. The applications therefore share parameters but have different feasible operating arrangements. Ownership, operating arrangements, and certification effort are held fixed within each experiment.

Separating access restrictions from strategic responses. The first comparison holds every network fixed and changes only the policy rule. In the second, every provider chooses its network assuming that competitors retain their baseline portfolio distributions. We then combine the chosen networks and jointly solve their prices. The third allows providers to respond to the policy distributions of their competitors. Thus all eight firms optimize in the latter two comparisons; what changes is the competing networks they expect to face.

Formally, let Q^O denote the baseline distribution. *Fixed networks* use Q^O in each regime. *Responses to unchanged rivals* use $Q^{r,u} = \bigotimes_f \widehat{\mathcal{B}}_f^r(Q_{-f}^O)$. *Equilibrium responses* use the policy-specific distribution Q^r computed on the common candidate sets. Prices are jointly solved for all providers in every market and profile, including in the unchanged-rivals comparison.¹⁹

5.1 EU public cloud: EU-wide eligibility and national fragmentation

Physical availability alone does not establish eligibility for public-sector cloud purchases. France’s SecNumCloud qualification attaches to a service offer and legal entity (Agence nationale de la sécurité des systèmes d’information, 2022). The European Commission’s *Cloud Sovereignty Framework*, applied in a 2026 procurement for EU institutions, assesses legal, operational, and technological control (European Commission, 2026a). The experiment isolates geographic supply eligibility within this broader institutional setting.

Under EU-wide eligibility, qualifying origins can serve regulated public customers throughout the model EU. Under national fragmentation, they can serve only domestic regulated demand. With the same regulated expenditure share $\omega = 0.18$, the eligibility

¹⁹These are static comparisons, not a temporal adjustment path. Individual one-firm effects need not add to their combined effect. The hat denotes the candidate-set approximation to the response operator; see Appendix F.

rules are

$$e_{in}^E = \mathbf{1}\{i \in \text{EU12}\}, \quad e_{in}^N = \mathbf{1}\{i = n\}, \quad n \text{ a regulated EU segment.}$$

The baseline imposes no geographic eligibility restriction on these customers; they use the ordinary global supply network. Unregulated demand retains its original eligibility in both policy regimes. National minus EU-wide thus changes the geographic scope of eligibility for the same customers and expenditure. National fragmentation is a scenario for sensitive public demand, not a general EU localization mandate: non-personal-data rules prohibit national localization subject to a proportionate public-security exception (European Parliament and Council of the European Union, 2018, Article 4). The experiment does not value security benefits or model ownership tests, sovereignty grades, or procurement auctions.

For comparison h , define

$$G_h = Y_N(Q^{N,h}) - Y_E(Q^{E,h}).$$

Prices are solved for every market and profile before aggregation. For a group of customer markets, Y is the expenditure-weighted expected log price index, and the percentage effect is $100[\exp(G_h) - 1]$. The difference between equilibrium responses and fixed networks measures total portfolio adjustment. The difference between equilibrium responses and responses to unchanged rivals measures the additional effect of changed competitor portfolios. Both EU-wide and National responses use the unrestricted baseline Q^O as the unchanged-rival benchmark. Their comparison therefore measures the difference in strategic adjustment between the two eligibility regimes, not adjustment from the EU-wide equilibrium to National.

Similar regulated prices, different international adjustment. Under equilibrium responses, residual suppliers account for 90.16 percent of regulated expenditure under National, compared with 33.73 percent under EU-wide and 12.50 percent in the baseline. Their supply is held fixed and remains eligible. The network results therefore describe adjustment by the eight strategic providers, not the entire industry; residual eligibility is examined below.

Table 4: National fragmentation: core networks, prices, and competitor responses

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
<i>Core networks</i>			
World core presences	0.00	-6.20	-4.11
EU core presences	0.00	-3.35	-2.79
Non-EU core presences	0.00	-2.84	-1.32
<i>Prices</i>			
Regulated price (%)	27.49	27.82	27.88
EU private price (%)	0.00	0.51	0.38
Non-EU price (%)	0.00	0.38	0.24

Notes: National minus EU-wide under the network comparisons defined in Section 5. Prices are solved jointly for each portfolio profile before aggregation. Core counts are expected provider-country presences. Residual supply remains locally eligible with a fixed price-index contribution. Table 14 reports the associated eligibility and residual-share measures.

Table 4 shows that competitor responses matter much more for the global network response than for the regulated-price effect. Responses to unchanged rivals produce 6.20 fewer worldwide core presences under National than under EU-wide. Equilibrium responses reduce that gap to 4.11, an attenuation of about one third. The regulated-price effects are much closer: 27.82 and 27.88 percent. Evaluating firms against unchanged rivals would therefore give a similar regulated-price effect while overstating the contraction of their multinational networks.

Most of this attenuation occurs outside the EU: the non-EU contraction falls from 2.84 to 1.32 core presences, whereas the EU contraction falls from 3.35 to 2.79. Tencent, OVH-cloud, and Oracle together account for about 82 percent of the worldwide attenuation. The effect of changed competitor portfolios is therefore concentrated in the international networks of a few providers, rather than being a uniform proportional adjustment. This is an accounting of the difference between the two response comparisons, not a decomposition into separate causal effects of each rival. A rival withdrawal changes the profits of both the network retaining a core and its best alternative without that core. If the former benefits more, retention becomes more attractive and contraction can be moderated; if the latter benefits more, the opposite can occur. The relevant mechanism is this change in relative network profitability.

Table 5: Core presence by eligibility regime

Core presences	Baseline	EU-wide	National
World	169.92	166.27	162.16
EU	52.54	53.34	50.55
Non-EU	117.38	112.93	111.61

Notes: Expected provider-country core presences of the eight modeled providers under the computed portfolio distribution of each regime (equilibrium responses). The baseline imposes no geographic eligibility restriction on regulated public customers; EU-wide and National are defined in Section 5.1. Table 4 reports the National minus EU-wide differences. Residual suppliers do not choose locations in the model.

Most of the regulated-price increase already occurs at fixed networks, where it is 27.49 percent. Portfolio adjustment adds only 0.39 percentage points, but it is the entire channel of transmission to other markets in this model. Unregulated markets retain their supply rules and expenditure, and pricing has no cross-market capacity or input-price channel. Their prices are therefore unchanged at fixed networks. With equilibrium responses, the EU-private price index rises by 0.38 percent and the non-EU index by 0.24 percent. Against unchanged rivals, the corresponding increases are 0.51 and 0.38 percent. Competitor responses moderate both the global contraction and its international price effects in this experiment; the general framework does not impose that direction of adjustment.

Domestic sourcing requirements need not attract more infrastructure. The contrast between EU-wide and National also appears relative to the baseline. Expected EU core presence rises from 52.54 to 53.34 under EU-wide, but falls to 50.55 under National (Table 5). EU-wide directs regulated demand toward EU origins while preserving each origin’s access to buyers across member countries. National favors domestic origins, but limits each eligible origin to its domestic regulated market. Access to that market need not justify a core when the location can no longer serve regulated buyers elsewhere in the EU. The relevant scale is the customer base accessible from an eligible origin, not the number of countries imposing a domestic-sourcing rule. Eligibility here concerns service origins: a local facility may still depend on a core in another country, subject to the maintained parent connection. EU-wide shifts core presence toward the EU while reducing the worldwide total. National instead reduces presence both inside and outside the EU.

All eight providers have fewer EU cores under National than under EU-wide (Figure 5). Outside the EU, AWS, Microsoft, and Google add a small number of presences while the

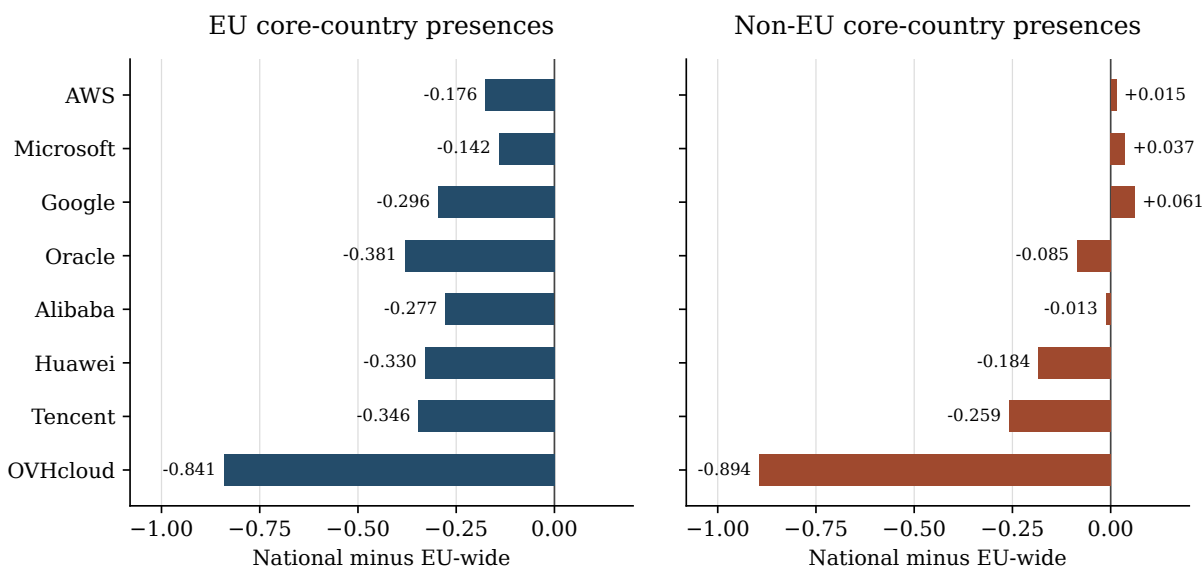


Figure 5: Where providers change their core networks

Notes: National minus EU-wide expected core-country presences under the computed portfolio distributions. EU and non-EU changes add to each firm’s worldwide change. These are provider-country presences, not building counts or installed capacity.

other providers contract. OVHcloud accounts for 42 percent of the worldwide count difference, but counts alone do not determine price effects. A core matters according to the markets it serves, the firm’s remaining supply, and its competitive position. The one-firm comparisons in Appendix Table 15 illustrate the distinction: OVHcloud removes more locations than Tencent but has a much smaller non-EU price effect. National also lowers all eight providers’ worldwide operating profits and profits net of location costs relative to EU-wide (Appendix Figure 7). Higher prices for regulated customers need not imply higher profits for the named providers.

Residual eligibility and starting geography. The large residual share under National makes its eligibility assumption quantitatively important. The calibration disciplines the level of residual supply, but does not establish which unmodeled suppliers would qualify.

Table 6 examines this assumption by repricing given networks. A larger residual contribution lowers the regulated-price effect, whereas making part of it ineligible under National raises it; across the rows the effect ranges from about 18 to 48 percent. The direction persists, but the magnitude depends on residual supply and its eligibility. Because portfolios are held fixed, these comparisons do not establish robustness of the network response.

Table 6: EU price sensitivity at fixed networks

Assumption	Regulated price	Residual share (%)	
	National–EU-wide (%)	EU-wide	National
<i>A. Residual supply and starting geography</i>			
Residual term $\times 0.5$	39.97	21.57	82.85
Baseline residual term	27.49	34.11	90.06
Residual term $\times 2$	17.86	49.04	94.56
Observed cores, baseline residual	22.02	40.92	90.80
<i>B. National eligibility of residual supply</i>			
National exposure $\phi = 0$	27.50	34.10	90.06
National exposure $\phi = 0.25$	35.97	34.10	87.43
National exposure $\phi = 0.5$	48.39	34.10	82.85

Notes: Panel A's first three rows hold the baseline portfolio distribution fixed and multiply residual price-index contribution in both policy regimes by 0.5, 1, or 2. Its final row uses observed cores with the same installed local topology and parameters. Panel B reduces the residual contribution only in National regulated markets, $b_n^N = (1 - \phi)b_n^E$, leaving EU-wide supply unchanged. It uses the baseline distribution recomputed with greater integration precision; the unexposed price effect differs from Panel A's baseline by less than 0.01 percentage points. All rows re-solve prices at fixed networks. These are conditional repricing comparisons, not recalibrations or portfolio equilibria under alternative residual supply. They assess magnitudes and starting geography without establishing the robustness of the portfolio response itself.

Starting geography also matters. With observed cores, the same installed local topology and the same parameters, the fixed-network price effect is 22.02 rather than 27.49 percent. This comparison changes the entire network; it does not isolate the baseline's excess EU core count. The residual assumption and geographic misfit therefore qualify the magnitudes, while the three response comparisons describe how strategic adjustment operates within the maintained model.

5.2 China: offshore supply costs and the multinational network

In mainland China, local operators supply AWS and Microsoft services through separate contracts and accounts (Appendix Table 12). A provider's branded service can therefore be available without a directly controlled core.

The second application changes the cost of serving mainland customers from offshore facilities. Supplier nationality does not determine exposure to this change: Chinese providers also serve mainland customers from offshore cores. A higher cost can therefore

favor a mainland location while making the same firm's offshore network less valuable. We examine partial easing and two tightening scenarios around the existing segmented environment:

$$\tau_{i,\text{CHN}}(d) = \tau_{i,\text{CHN}}(0) \exp\{d \mathbf{1}(i \neq \text{CHN})\}, \quad d \in \{-0.10, 0, 0.10, 0.25\}.$$

The offshore-to-China trade-cost factor falls by 9.52 percent under easing and rises by 10.52 or 28.40 percent under tightening. These changes are scenarios around existing offshore supply barriers. Domestic supply costs, other destinations, accessibility, expenditure, and location-cost and demand parameters remain unchanged.

The calculation keeps these partner-operated branded paths fixed rather than treating them as direct core choices. The five non-Chinese providers cannot add a directly controlled mainland core in this scenario; Alibaba, Huawei, and Tencent can choose both mainland and offshore cores. Foreign providers can continue offshore service to China.²⁰ The baseline distribution is recomputed under these arrangements with $d = 0$. All four scenarios use the same calibrated parameters and candidate portfolios. The baseline overstates the role of foreign branded suppliers relative to published Chinese market statistics. Appendix F.4 documents this mismatch and the difference in product coverage; the experiment is conditional on the maintained market composition and operating arrangements.

Similar domestic prices, different global responses. Table 7 uses the same three comparisons as the EU experiment. Under the larger tightening, the China price index rises by 19.37 percent at fixed networks and 20.25 percent with equilibrium responses. Responses to unchanged rivals give 20.72 percent. Yet their worldwide core contraction is 7.36, compared with 3.90 under equilibrium responses: allowing for changed rival portfolios reduces the contraction by about 47 percent. The non-China price increase correspondingly falls from 0.70 to 0.42 percent. As in the EU application, international price effects are zero at fixed networks because other markets' supply conditions and expenditure do not change directly. All eight providers retain more worldwide core presences under equilibrium responses than in the unchanged-rivals comparison for this tightening. Tencent,

²⁰These restrictions describe the maintained operating arrangements, not a general legal prohibition of foreign participation. Partner paths receive no private core-cost draws. The central accounting assigns their modeled branded operating profit to the provider; partnership formation and profit-sharing changes are outside the experiment.

Table 7: China: prices and core networks under easing and tightening

Network response	China price (%)	Other prices (%)	World cores (presences)	China cores (presences)
<i>Easing: trade cost -9.52%</i>				
Fixed networks	-7.84	0.00	0.00	0.00
Responses to unchanged rivals	-8.11	-0.14	+1.43	-0.03
Equilibrium responses	-8.05	-0.11	+0.96	-0.04
<i>Moderate tightening: trade cost +10.52%</i>				
Fixed networks	+7.89	0.00	0.00	0.00
Responses to unchanged rivals	+8.36	+0.24	-2.49	+0.07
Equilibrium responses	+8.17	+0.13	-1.21	+0.08
<i>Larger tightening: trade cost +28.40%</i>				
Fixed networks	+19.37	0.00	0.00	0.00
Responses to unchanged rivals	+20.72	+0.70	-7.36	+0.12
Equilibrium responses	+20.25	+0.42	-3.90	+0.16

Notes: All scenarios use common candidate sets and the same baseline reference, which retains existing segmentation. The three response comparisons are defined in Section 5. Prices are solved jointly for every profile; reported changes use expenditure-weighted expected log indices. Core counts are expected provider-country presences, excluding fixed partner paths. Residual supply is unexposed to the shock ($\phi = 0$); its share changes are reported in Table 16.

OVHcloud, and Oracle account for the largest differences. Thus the aggregate attenuation reflects changes in the networks of both Chinese and foreign providers, even though Chinese providers account for most of the remaining net contraction.

Competitor responses also moderate global contraction under the smaller tightening and global expansion under easing. Easing lowers prices both in China and abroad. Attenuation in both directions is a result of these experiments, not a general implication of strategic portfolio choice.

Greater domestic presence need not imply a larger worldwide network or higher profit. Under the larger tightening, Alibaba’s mainland-core probability rises by 10.18 percentage points, Huawei’s by 3.43, and Tencent’s by 2.21. Their worldwide networks nevertheless contract by 0.68, 1.20, and 1.79 core-country presences (Figure 6). These three providers

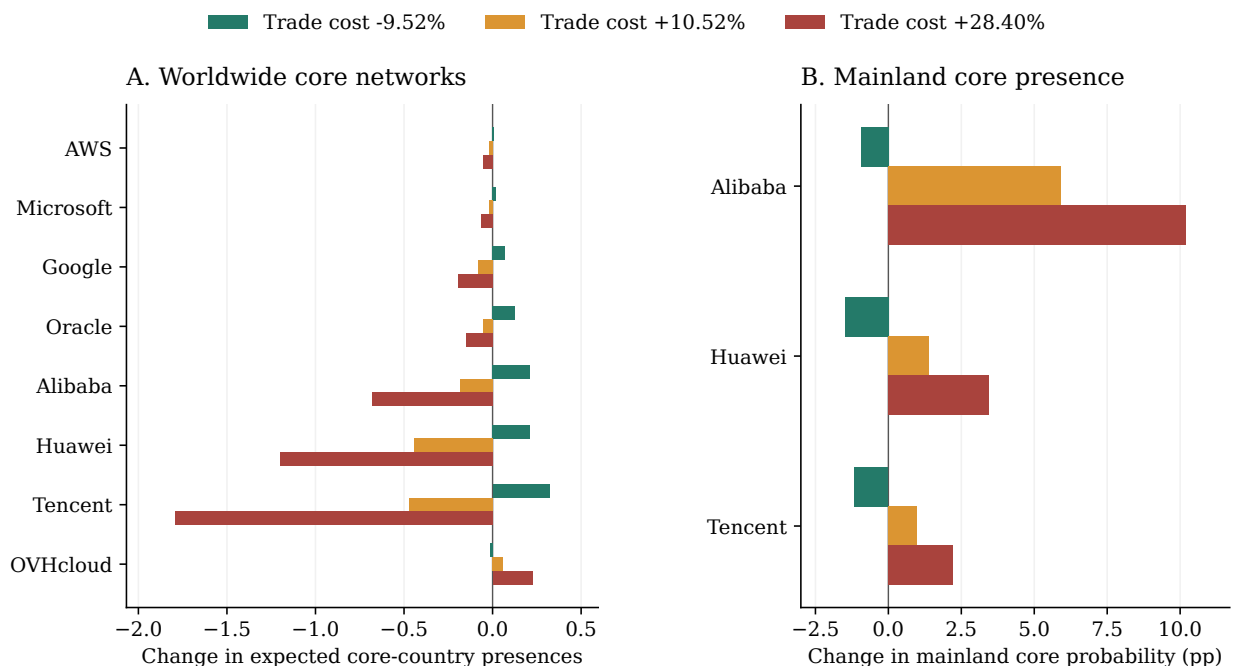


Figure 6: China frictions: mainland presence and the global portfolio

Notes: Equilibrium responses relative to the baseline under maintained operating arrangements. Panel A reports worldwide core-country changes; Panel B reports mainland-core probability changes for the three providers that can select such a core. Foreign direct-core probabilities are fixed at zero; AWS and Microsoft retain partner paths that are excluded from direct-core counts.

account for about 94 percent of the net worldwide contraction. Across all eight firms, mainland presence rises by 0.158 while offshore presence falls by 4.06. The response is therefore not a one-for-one relocation to China.

The three Chinese providers' worldwide operating profits fall by 7.43, 12.19, and 15.86 percent, respectively; their profits net of location costs also fall (Appendix Figure 7). They serve mainland customers from offshore locations as well as domestic ones, so a restriction on offshore supply raises their own delivery costs. Greater domestic presence is consistent with a smaller international network and lower worldwide profit. Easing reverses this pattern: their mainland probabilities decline while worldwide networks expand and operating profits rise by about 2–3 percent.

Supply origins and the incidence across providers. The central calculation leaves residual supply unexposed to the offshore friction. Its China share rises from 9.78 to 20.39 percent under larger tightening, while the three Chinese providers' combined total-market

share falls by 4.56 percentage points. This incidence depends on where the residual supply originates. Exposing its entire contribution to the same friction, while repricing the central policy portfolios, raises the China price effect from 20.25 to 25.63 percent and changes the Chinese providers' share effect to approximately +0.09 percentage points (Appendix Table 17). Provider nationality alone therefore does not determine who gains market share from a border restriction. Because this comparison reprices without reoptimizing portfolios, the network and profit responses above remain conditional on the central assumption (Appendix F.4).

6. CONCLUSION

Geographic restrictions change the customer base of facilities throughout a multinational network. In the cloud applications, competitor responses substantially moderate worldwide network contraction while leaving price effects in directly affected markets largely unchanged. Domestic presence and worldwide network size can also move in opposite directions. EU-wide eligibility expands aggregate EU core presence relative to the baseline without geographic eligibility restrictions, whereas national eligibility reduces it: a core serving customers throughout the EU need not be profitable when restricted to domestic regulated demand. In China, higher offshore costs raise Chinese providers' mainland presence while shrinking their worldwide networks and operating profits.

To analyze these responses when a location's net return can change sign as networks expand, we optimize complete portfolios using profit bounds that tie market benefits and location costs to the same portfolio. These bounds eliminate groups of unexamined networks by limiting what the same facilities can supply and what they cost. They allow endogenous portfolio size and report upper bounds on potential gains from deviations to any feasible portfolio. Response searches combined with distribution updates connect network optimization to equilibrium analysis; the enumeration comparisons measure the computational gains separately from equilibrium approximation accuracy.

The quantitative magnitudes depend on the maintained geography, residual supply, operating arrangements, and numerical approximation documented in Appendix F. Within that scope, the experiments show why provider nationality, facility location, and customer location must be distinguished. A restriction intended to favor domestic supply

can alter the profitability of networks abroad and the prices paid by customers outside the regulated market.

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APPENDIX

The appendix contains the proofs, computational procedures, supporting evidence, data conventions, and policy diagnostics.

A. EQUILIBRIUM AND PRICING PROOFS

A.1 Existence and contingent cutoffs

Proof of Proposition 2. The state space $\mathcal{Q} = \prod_f \Delta(\mathcal{X}_f)$ is nonempty, compact, and convex, and each portfolio's interim value is continuous in $\mathcal{Q}$. Distinct portfolios have distinct private-cost loading vectors, so their indifference set is an affine hyperplane of probability zero. If $\mathcal{Q}^k \rightarrow \mathcal{Q}$, the best-response indicator therefore converges for almost every type. Although a firm's choice is discrete, the probability of switching tends to zero as the payoff change vanishes. Dominated convergence gives continuity of $\mathcal{B} : \mathcal{Q} \rightarrow \mathcal{Q}$, and Brouwer's theorem gives a fixed point. This argument applies to any continuous interim-value tables; it does not require multilinearity. $\square$

Inclusion after reoptimizing the remaining network. Fix $\mathcal{Q}_{-f}$ and the other costs $\varepsilon_{f,-j}$, and put $v_f(S) = V_f(S; \mathcal{Q}_{-f})$. For a location j whose inclusion varies across $\mathcal{X}_f$, define

$$V_{fj}^{\text{in}} = \max_{\substack{S \in \mathcal{X}_f \\ j \in S}} \left\{ v_f(S) - \sigma_\varepsilon \sum_{k \in S \setminus \{j\}} \ell_{fk} \varepsilon_{fk} \right\}, \quad V_{fj}^{\text{out}} = \max_{\substack{S \in \mathcal{X}_f \\ j \notin S}} \left\{ v_f(S) - \sigma_\varepsilon \sum_{k \in S} \ell_{fk} \varepsilon_{fk} \right\}.$$

Lemma A.1 (Contingent inclusion cutoff). *Outside the null tie set, the optimal portfolio contains j if and only if*

$$\varepsilon_{fj} < \bar{\varepsilon}_{fj}(\varepsilon_{f,-j}; v_f) = \frac{V_{fj}^{\text{in}} - V_{fj}^{\text{out}}}{\sigma_\varepsilon \ell_{fj}}. \quad (\text{A.1})$$

The two maximizing networks can contain different other locations. The cutoff is therefore a reservation cost after reoptimization, not a value assigned to j independently of the remaining costs.

For the proof, condition on $\varepsilon_{f,-j}$. All include- j portfolios have slope $-\sigma_\varepsilon \ell_{fj}$ in ε_{fj} , and all exclude- j portfolios have slope zero. Their optimized values therefore differ by $V_{fj}^{\text{in}} - V_{fj}^{\text{out}} - \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$, which proves the cutoff formula outside the null tie set. Coordinates constant on the feasible family are not decision margins.

A.2 Portfolio-response bounds

Let $P_f(v)$ be the optimal portfolio law at value table v , and let $\mathcal{J}_f$ contain the coordinates varying across $\mathcal{X}_f$. Put $\gamma_{fj} = \bar{g}_{fj}/\ell_{fj}$. For a payoff perturbation h , define

$$C_f(h) = \min_{c \geq 0} \left\{ \sum_{j \in \mathcal{J}_f} \gamma_{fj} c_j : |h(S) - h(T)| \leq \sum_{j \in S \Delta T} c_j \quad (S, T \in \mathcal{X}_f) \right\}. \quad (\text{A.2})$$

This covers changes in relative portfolio payoffs by weights on the locations where the alternatives differ.

Lemma A.2 (Probability flow across portfolio boundaries). *Under Assumption 1, C_f is a seminorm and*

$$d_{\text{TV}}(P_f(v + h), P_f(v)) \leq C_f(h)/\sigma_\varepsilon. \quad (\text{A.3})$$

It is no larger than the include-exclude bound $\|h\|_{\partial f} = \sum_j \gamma_{fj} \max_{S \ni j, T \not\ni j} |h(S) - h(T)|$.

For rival portfolios A, B and other rivals' portfolios U , set $\Delta r_{A,B,U}^{fg}(S) = r_f(S, A, U) - r_f(S, B, U)$ and

$$K_{fg}^\# = \max_{A,B,U} C_f(\Delta r_{A,B,U}^{fg}), \quad K_{ff}^\# = 0. \quad (\text{A.4})$$

The comparison proved below gives

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sigma_\varepsilon^{-1} \sum_{g \neq f} K_{fg}^\# d_{\text{TV}}(Q_g, Q'_g). \quad (\text{A.5})$$

Total variation need not equal the fraction of types switching under a common draw of private costs.

A.3 Proof of the probability-flow lemma

Suppress the firm index. Remove constant coordinates: their costs cancel from every comparison, and conditional density caps on the remaining coordinates survive marginalization by conditional expectation. Write $z_S \in \{0, 1\}^m$ for the distinct remaining incidence vectors and $L = \text{diag}(\ell_j)$. A singleton family has zero sensitivity.

The cover exists because $c_j = \text{osc}(h)$ is feasible. Truncating any component at that value preserves feasibility, so the minimum is attained. Scaling covers and adding them prove homogeneity and the triangle inequality. The include-exclude maximum for coordinate j covers every pair that differs at j , proving the stated comparison.

Take a feasible cover c and interpolate $v_t = v + th$, $0 \leq t \leq 1$. Couple choices using the same cost vector throughout. If N counts changes in the maximizing portfolio along this segment, then

$$d_{TV}(P(v+h), P(v)) \leq \Pr\{\text{endpoint choices differ}\} \leq \mathbb{E}N.$$

There are finitely many switches almost surely: the choice is the upper envelope of finitely many affine functions of t . A switch between S, T requires $d_{ST} = h(S) - h(T) \neq 0$. Allocate its unit weight across changed coordinates as $\omega_{j,ST} = c_j / \sum_{k \in S \Delta T} c_k$. The cover implies $\omega_{j,ST} |d_{ST}| \leq c_j$. These weights allocate one whole-portfolio switch across the locations that change, avoiding multiple counts of the same switch. The c_j cover relative payoff changes; they are not individual locations' marginal returns.

First let the density p be smooth with $p(e) \leq \bar{g}_j p_{-j}(e_{-j})$. For each switching pair and $j \in S \Delta T$, the tie equation maps t to e_j with absolute derivative $|d_{ST}| / (\sigma_\epsilon \ell_j)$. At fixed (t, e_{-j}) , the best include- j and exclude- j portfolios are unique almost everywhere and meet at the single cutoff $\bar{e}_j(t, e_{-j})$ from Lemma A.1. Hence summing the weighted crossings and changing variables gives

$$\mathbb{E}N \leq \frac{1}{\sigma_\epsilon} \sum_j \frac{c_j}{\ell_j} \int_0^1 \int p(\bar{e}_j(t, e_{-j}), e_{-j}) de_{-j} dt \leq \frac{1}{\sigma_\epsilon} \sum_j \frac{c_j \bar{g}_j}{\ell_j}.$$

Within-group ties are nontrivial hyperplanes in the other costs. Triple top-choice ties have zero crossing measure because three distinct binary incidence vectors cannot be collinear. Thus neither creates an additional switching term. For an arbitrary absolutely continuous density, convolve with a nonnegative product kernel φ_ϵ whose coordinate factors integrate to one. Each conditional cap is preserved because

$$p \leq \bar{g}_j p_{-j} \implies p * \varphi_\epsilon \leq \bar{g}_j (p_{-j} * \varphi_{\epsilon, -j}) = \bar{g}_j (p * \varphi_\epsilon)_{-j}.$$

Convergence in L^1 gives total-variation convergence of the distributions induced by each fixed endpoint choice map, passing the inequality to the original choices. Minimize over covers to obtain the result. Bounds valid throughout the payoff segment can replace the global coordinate caps.

Boundary weights for smooth densities. For later use, let $\Gamma_{ST}(t)$ be a shared top-choice facet and define $w_{ST}(t) = \int_{\Gamma_{ST}(t)} p(e) / \|L(z_S - z_T)\|_2 d\mathcal{H}^{m-1}(e)$. Projection onto e_{-j} has Jacobian $\ell_j / \|L(z_S - z_T)\|_2$. The same single-cutoff argument therefore gives

$$\sum_{\{S,T\}: j \in S \Delta T} w_{ST}(t) = \ell_j^{-1} \int p(\bar{e}_j(t, e_{-j}), e_{-j}) de_{-j} \leq \bar{g}_j / \ell_j. \quad (\text{A.6})$$

This is the boundary-mass bound used for Gaussian costs in Appendix G.1.

A.4 Independent normal costs

When all subsets are feasible, put $D_j(h) = \max_{S:j \notin S} |h(S \cup \{j\}) - h(S)|$. Adjacent pairs require $c_j \geq D_j(h)$; toggling each differing coordinate once shows that these values cover every pair. Therefore

$$C_f(h) = \sum_{j \in \mathcal{J}_f} \gamma_{fj} D_j(h). \quad (\text{A.7})$$

For independent standard normal shocks and any feasible family, write $L_f = \text{diag}(\ell_{fj})$ and $m_a(S) = \sum_{j \in S \cap \mathcal{J}_f} a_j$. Split the payoff change as $h = m_a + (h - m_a)$. The modular part m_a adds the same return a_j to location j regardless of which other locations the firm chooses. In the choice problem it is equivalent to a cost shift:

$$m_a(S) - \sigma_\varepsilon \sum_{j \in S \cap \mathcal{J}_f} \ell_{fj} \varepsilon_{fj} = -\sigma_\varepsilon \sum_{j \in S \cap \mathcal{J}_f} \ell_{fj} \left(\varepsilon_{fj} - \frac{a_j}{\sigma_\varepsilon \ell_{fj}} \right).$$

This equivalence bounds the response to a payoff change; it does not make primitive private costs depend on rivals. It motivates the bound

$$C_f^N(h) = \inf_a \left\{ \frac{\|L_f^{-1}a\|_2}{\sqrt{2\pi}} + C_f(h - m_a) \right\}, \quad \gamma_{fj} = \frac{1}{\ell_{fj}\sqrt{2\pi}}. \quad (\text{A.8})$$

This seminorm satisfies $C_f^N \leq C_f$ and the probability bound (A.3) with C_f^N in place of C_f . To prove it, first compare v with $v + m_a$. The identity translates the normal costs by $-L_f^{-1}a/\sigma_\varepsilon$ without changing the choice rule. Applying a fixed choice rule cannot increase total variation, and the distance between $N(0, I)$ and $N(d, I)$ is $2\Phi(\|d\|_2/2) - 1 \leq \|d\|_2/\sqrt{2\pi}$. Next compare $v + m_a$ with $v + h$ using the probability-flow lemma for the residual $h - m_a$. The triangle inequality and minimization over a give the stated bound. Scaling and combining decompositions establish the seminorm property; $a = 0$ establishes domination. Replacing C_f by C_f^N in (A.4) defines $K^N \leq K^\#$.

On the full cube let $\underline{d}_j, \bar{d}_j$ be the minimum and maximum marginal perturbations, and put $b_j = (\bar{d}_j + \underline{d}_j)/2$, $r_j = (\bar{d}_j - \underline{d}_j)/2$. Then

$$C_f^N(h) = \frac{1}{\sqrt{2\pi}} \left[\|L_f^{-1}b\|_2 + \sum_j \frac{r_j}{\ell_{fj}} \right]. \quad (\text{A.9})$$

The midpoint b_j isolates a common location-return shift; the half-width r_j bounds how much that return varies with the rest of the portfolio. To see this, $D_j(h - m_a) = r_j + |b_j - a_j|$ and $\|L_f^{-1}b\|_2 \leq \|L_f^{-1}a\|_2 + \|L_f^{-1}(b - a)\|_1$, with equality in the minimizing bound at $a = b$. Enclosing the marginal extrema by wider intervals gives a valid upper bound. This formula solves the stated decomposition problem, not every possible sensitivity bound.

A.5 Equilibrium comparison and uniqueness

Proposition A.1 (Equilibrium comparison). *Suppose a nonnegative matrix M bounds response distances componentwise on $\mathcal{U} \subseteq \mathcal{Q}$: $d(\mathcal{B}(Q), \mathcal{B}(Q')) \leq Md(Q, Q')$, where $d_f(Q, Q') = d_{TV}(Q_f, Q'_f)$. If $\mathcal{U}$ contains every equilibrium and $\rho(M) < 1$, the equilibrium Q^* is unique. For any $\tilde{Q} \in \mathcal{U}$, its exact-response residual $e_f = d_{TV}(\tilde{Q}_f, \mathcal{B}_f(\tilde{Q}_{-f}))$ satisfies*

$$d(\tilde{Q}, Q^*) \leq (I - M)^{-1}e. \quad (\text{A.10})$$

Iteration converges from $\mathcal{U}$ if that set is invariant, or from all of $\mathcal{Q}$ if $\mathcal{B}^L(Q) \subseteq \mathcal{U}$ for some finite L .

Proof. Any difference between two equilibria must be reproduced by firms' responses to one another. Two fixed points have distance vector $d \leq Md \leq M^k d \rightarrow 0$. For an approximate point, $d \leq e + Md$; iterating and using $(I - M)^{-1} = \sum_{k \geq 0} M^k \geq 0$ gives the residual bound. Each term $M^k e$ allows another round of propagation of the initial residual. Iterates in $\mathcal{U}$ obey $d^{t+1} \leq Md^t$. Invariance keeps them there; finite-step containment does so from time L onward because $\mathcal{B}^t(Q) = \mathcal{B}^L(\mathcal{B}^{t-L}(Q))$. Existence is established above. $\square$

To prove (A.5), telescope the product of rival laws one marginal at a time. The positive and negative parts of $Q'_g - Q_g$ each have mass $\delta_g = d_{TV}(Q_g, Q'_g)$. Coupling their normalized parts by $\pi_g(A, B)$ expresses the associated payoff change as

$$h_g(S) = \delta_g \sum_{A, B, U} \pi_g(A, B) \omega_g(U) \Delta r_{A, B, U}^{fg}(S),$$

where ω_g is the hybrid law of the other rivals. Homogeneity and subadditivity give $C_f(h_g) \leq \delta_g K_{fg}^\#$. Summing and applying the probability-flow bound proves the comparison; the same argument gives K^N under normal costs. Take $\mathcal{U} = \mathcal{Q}$ and $M = K^\#/\sigma_\varepsilon$ in the proposition to prove Theorem 1. In particular, $a = (I - M)^{-1}\mathbf{1} \gg 0$ makes the response map a contraction in $\max_f d_f/a_f$, since $Ma = a - \mathbf{1} < a$. The spectral radius is the infimum of the comparison constants over positive firm weights (Horn and Johnson, 2013, Chapter 8). A finite-menu or sampled-type residual needs a separate approximation allowance before it can be used as the exact residual in (A.10).

A.6 Conditional AB pricing and derivatives

Fix $\sigma > \eta > 1$, an active set, and a market, suppressing its index. Let $y_f = \psi_f p_f^{1-\sigma}$ and $D_f = b + \sum_{g \neq f} y_g$, where $b \geq 0$ is fixed residual price-index contribution. For a given $D_f > 0$, hold $\xi_f = \psi_f c_f^{1-\sigma} > 0$ fixed. A scalar pricing response solves

$$y = \xi_f \mu \left(\frac{y}{y + D_f} \right)^{1-\sigma}.$$

A higher candidate y raises the firm's share and markup, lowering the right-hand side while the left-hand side rises. The markup bounds keep the right-hand side positive and bounded, so the curves meet exactly once. To differentiate this response, put $s_f = y/(y + D_f)$ and

$$\beta(s) = (\sigma - 1) \frac{d \log \mu(s)}{ds} = \frac{(\sigma - 1)(\sigma - \eta)\mu(s)}{\sigma\eta} > 0, \quad \beta_f = \beta(s_f).$$

The log equation has derivative $1 + s_f(1 - s_f)\beta_f > 0$ with respect to $\log y$, so the unique response $y_f = \mathcal{Y}_f(D_f)$ is smooth.

Differentiating $\log y_f = \log \xi_f + (1 - \sigma) \log \mu(s_f)$ gives $d \log y_f = -\beta_f ds_f$. Combining this with $ds_f = s_f(1 - s_f)(d \log y_f - d \log D_f)$ gives $d \log y_f = q_f(d \log D_f - d \log y_f)$, where $q_f = \beta_f s_f(1 - s_f)$. Hence

$$\tilde{\kappa}_f = \frac{\partial \log y_f}{\partial \log D_f} = \frac{q_f}{1 + q_f} < 1. \quad (\text{A.11})$$

An increase in rival contribution lowers the firm's share and induces a lower markup. The resulting recovery in its own contribution limits the markup adjustment. Thus its contribution responds less than proportionally to D_f , holding its own competitiveness fixed. Here s_f is evaluated at the solved scalar response $\mathcal{Y}_f(D_f)$. Define $x_f = \log y_f$ and $\Phi_f(x) = \log \mathcal{Y}_f(b + \sum_{g \neq f} e^{x_g})$. Its Jacobian has

$$\Lambda_{fg} = \tilde{\kappa}_f \frac{y_g}{D_f} \quad (g \neq f), \quad \Lambda_{ff} = 0. \quad (\text{A.12})$$

In this Jacobian, $y_g = e^{x_g}$ for rivals, while $\tilde{\kappa}_f$ uses the solved scalar response, not the share implied by the input x_f . Cross-market entries are zero. Since $\sum_{g \neq f} y_g/D_f \leq 1$, $\beta_f \leq \bar{\beta} = (\sigma - 1)(\sigma - \eta)/[\sigma(\eta - 1)]$, and $s_f(1 - s_f) \leq 1/4$,

$$\|\Lambda(x)\|_\infty \leq \bar{\Lambda}_{AB} = \frac{\bar{\beta}/4}{1 + \bar{\beta}/4} < 1. \quad (\text{A.13})$$

The map is defined on the Euclidean space of log contributions x ; the mean-value inequality and Banach's theorem give a unique fixed point. If $b = 0$ and only one firm is active, $s_f = 1$ and $p_f = \mu(1)c_f$ directly. With $b > 0$ and one named firm, its scalar response faces the constant $D_f = b$. An empty active set has no named pricing variables; if residual supply is also absent, no aggregate price index is asserted. These cases prove existence and uniqueness for every portfolio profile in Proposition 1.

On a fixed active set, $I - \Lambda$ is invertible and the implicit-function theorem gives smooth dependence on positive costs and accessibility. Moreover $\|(I - \Lambda)^{-1}\|_\infty \leq (1 - \bar{\Lambda}_{AB})^{-1}$. At $(\sigma, \eta) = (5, 2)$ the uniform modulus is 0.375. This bounds conditional pricing feedback. Uniqueness of the portfolio equilibrium requires a separate condition.

Lemma A.3 (Equilibrium share and profit derivatives). *On a nonempty fixed active set I , write $b \geq 0$ for the residual contribution to the price-index sum and set $a_0 = 1 - 1/\sigma$, $b_0 = 1/\eta - 1/\sigma$. For log pre-markup*

scores $t_i = \log \xi_i$, the equilibrium system is

$$Y s_i = \xi_i (a_0 - b_0 s_i)^{\sigma-1}, \quad s_0 = b/Y, \quad s_0 + \sum_{i \in I} s_i = 1. \quad (\text{A.14})$$

Define

$$r(s) = \frac{s}{1 + \beta(s)s} = \frac{s(a_0 - b_0 s)}{a_0 + (\sigma - 2)b_0 s}, \quad W = s_0 + \sum_i r(s_i), \quad \omega_i = r(s_i)/W.$$

The exact share derivatives are

$$J_{ik} = r(s_i)(\mathbf{1}\{i = k\} - \omega_k), \quad J_{0k} = -s_0 \omega_k, \quad J_{0k} + \sum_i J_{ik} = 0. \quad (\text{A.15})$$

With $\underline{\kappa}_{AB} = (a_0 - b_0)/[a_0 + (\sigma - 2)b_0] > 0$,

$$\underline{\kappa}_{AB} s_i (1 - s_i) \leq J_{ii} \leq s_i (1 - s_i). \quad (\text{A.16})$$

Hold rival scores and b fixed. For $R \geq 0$ and $G(t_i) = R[s_i/\sigma + b_0 s_i^2]$, if $[s^L, s^U]$ encloses the own share throughout a fixed-active-set score interval, then

$$G'(t_i) \geq \underline{\kappa}_{AB} R \min\{u(s^L), u(s^U)\}, \quad u(s) = (1/\sigma + 2b_0 s)s(1 - s). \quad (\text{A.17})$$

Proof. Logarithmic differentiation of (A.14) gives $ds_i = r(s_i)(dt_i - d \log Y)$ and $ds_0 = -s_0 d \log Y$. The own-score gain is net of the common price-index adjustment; $r(s_i) \leq s_i$ captures the dampening effect of the firm's markup response. Adding these equations yields $d \log Y = \sum_i \omega_i dt_i$, proving (A.15). The denominator W is positive on a nonempty active set, including a monopoly without residual supply.

The ratio $r(s)/s = (a_0 - b_0 s)/[a_0 + (\sigma - 2)b_0 s]$, continuously extended at zero, has derivative $-(\sigma - 1)a_0 b_0/[a_0 + (\sigma - 2)b_0 s]^2 < 0$. Its denominator remains positive when $\sigma < 2$, because $a_0 + (\sigma - 2)b_0 = (a_0 - b_0) + (\sigma - 1)b_0 > 0$. Thus $\underline{\kappa}_{AB} s \leq r(s) \leq s$. Put $B_i = s_0 + \sum_{g \neq i} r(s_g)$. Then $\underline{\kappa}_{AB}(1 - s_i) \leq B_i \leq 1 - s_i$ and

$$J_{ii} = \frac{r(s_i)B_i}{r(s_i) + B_i}. \quad (\text{A.18})$$

Monotonicity of $xy/(x + y)$ in nonnegative x, y gives (A.16); both bounds are zero for a monopoly without residual supply. Finally, $G' = R(1/\sigma + 2b_0 s_i)J_{ii} \geq \underline{\kappa}_{AB} R u(s_i)$. The derivative $u'(s) = 1/\sigma + (4b_0 - 2/\sigma)s - 6b_0 s^2$ has one negative and one positive root, is positive at zero and negative at one. Hence u increases then decreases on $[0, 1]$; its minimum on an interval is attained at an endpoint. This proves (A.17), including the zero bounds when $R = 0$ or an endpoint is zero or one. $\square$

Residual price-index contribution is fixed in this calculation, but its share varies. The no-residual formulas follow by setting $s_0 = 0$. Derivatives on a fixed active set do not justify integrating

through an entry or exit boundary. With positive residual supply, own profit extends continuously to zero as its score vanishes.

A.7 Location interactions after repricing

Use the notation of Lemma A.3, and assume some competing supply, so $0 < s_i < 1$. Put $a_0 = 1 - 1/\sigma$, $b_0 = 1/\eta - 1/\sigma$, and $h(s) = s/\sigma + b_0 s^2$, so operating profit is $Rh(s_i)$. If $\xi_i = \omega_i x^{\nu s}$, its curvature is

$$\Pi_i''(x) = \frac{R\nu s h'(s_i) D_i}{x^2} \underbrace{\left[\nu s \left\{ \frac{h''(s_i)}{h'(s_i)} D_i + E_i \right\} - 1 \right]}_{\Xi_i}. \quad (\text{A.19})$$

The factor outside brackets is positive. To derive the responses, put $r_i = r(s_i)$, $\omega_i = r_i/W$, $D_i = r_i(1 - \omega_i)$, and

$$T_{-i} = \frac{s_0 + \sum_{k \neq i} r_k r'(s_k)}{W - r_i}.$$

This quantity averages the pricing responses of the competing suppliers; the fixed residual contributes weight s_0 and derivative one. In contrast, its market share is endogenous. Writing $t_i = \log \xi_i$, direct logarithmic differentiation of $D_i = r_i(W - r_i)/W$ gives

$$E_i \equiv \frac{\partial \log D_i}{\partial t_i} = (1 - \omega_i)^2 r'(s_i) - \omega_i^2 T_{-i}. \quad (\text{A.20})$$

Indeed, $\partial r_i / \partial t_i = r'(s_i) D_i$ and $\partial(W - r_i) / \partial t_i = -\omega_i(W - r_i) T_{-i}$; substituting into the log derivative proves the expression without any sign assumption on r' . If $\xi_i = \omega_i x^{\nu s}$, then $\Pi_i'(x) = R\nu s h'(s_i) D_i / x$. Another derivative gives (A.19). At $\eta = \sigma$, $r(s) = s$, $W = 1$, $E_i = 1 - 2s_i$, and $h'' = 0$, giving the stated CES expression.

The rows of Table 1 correspond to admissible pricing equilibria: choose $Y = 1$, $b = s_0$, and $\xi_k = s_k / (a_0 - b_0 s_k)^{\sigma-1}$. For a positive continuous supply mapping, the four rows give $\Xi_i = -0.13065$, 0.08322 , -0.00705 , and 0.09511 , respectively, under the table's parameters. Local interactions between two supply increments have the sign of Π_i'' for sufficiently small increments. For arbitrary finite increments, the fundamental theorem of calculus instead gives

$$\Pi_i(x + d_j + d_k) - \Pi_i(x + d_j) - \Pi_i(x + d_k) + \Pi_i(x) = \int_0^{d_j} \int_0^{d_k} \Pi_i''(x + u + v) dv du.$$

A nonsmooth eligibility or parent-activation mapping requires direct portfolio comparisons; the local derivative alone does not cover it.

Proof of Proposition 3. Let $g_d(x) = \Pi_i(x + d) - \Pi_i(x)$ be the operating-profit gain from the same location, contributing d , when existing supply is x . We compare this gain with one fixed location cost at three levels of existing supply. Operating profit $\Pi_i(x)$ is strictly increasing for $x > 0$, since

$h' > 0$ and $D_i > 0$, and is bounded above by R/η . Bounded markups and positive competing supply imply $s_i(x) = O(x^{v_s})$ as $x \downarrow 0$, hence $\Pi_i(x) = O(x^{v_s})$. Choose any $x_M > 0$. As $d \downarrow 0$,

$$\frac{g_d(d)}{d} \longrightarrow 0, \quad \frac{g_d(x_M)}{d} \longrightarrow \Pi'_i(x_M) > 0.$$

Choose a small $d < x_M$ such that $g_d(d) < g_d(x_M)/2$, and set the cost of location j to $c_j = g_d(x_M)/2 > 0$. Hold both d and c_j fixed. Because Π_i is increasing and bounded, $g_d(x) \rightarrow 0$ as $x \rightarrow \infty$. Choose $x_R > x_M$ with $g_d(x_R) < c_j$. A base supply $x_0 = d$ and contributions $d_j = d$, $d_k = x_M - d$, and $d_h = x_R - x_M$ give the three strict signs. The other locations' additive costs cancel from these comparisons. The proof does not require a global S-shape: the small-supply limit and bounded profit suffice. It also applies at $\eta = \sigma$. $\square$

B. VALID PORTFOLIO-COMPLETION BOUNDS

B.1 Cardinality conditioning and pricing envelopes

Proof of Proposition 4. For any completion $T = S \cup H$, $H \subseteq U$, $|H| = k$, each additive statistic satisfies

$$z_{\varsigma m}(T) = z_{\varsigma m}(S) + \sum_{j \in H} a_{\varsigma m j} \leq z_{\varsigma m}(S) + \text{Top}_k(a_{\varsigma m}).$$

Coordinatewise monotonicity of the continuation envelope therefore bounds its operating profit. Its cost is at least $C(S) + \text{Bot}_k(C_U)$, including when costs are negative. Average over rival states with their nonnegative probabilities, subtract this cost lower bound, and maximize over attainable k to obtain (3.7). Relaxing additional feasibility restrictions preserves an upper bound. $\square$

Under CES, nonnegative service and accessibility contributions and nonnegative accessibility loadings make own competitiveness increasing in the statistics. The share is increasing in competitiveness and variable profit is $R_n s_{fn}/\sigma$. Under AB, variable profit is $R_n h(s_{fn})$, where $h(s) = s/\sigma + (1/\eta - 1/\sigma)s^2$ is increasing for $\sigma > \eta > 1$. A valid increasing share envelope therefore suffices.

A markup-based share envelope. Fix a rival state ς . Let $\bar{\psi}_{fn}$ and $\underline{c}_{fn}$ bound own accessibility above and delivered cost below for every completion. For parent-dependent service, these bounds must cover every supported path. Put

$$\bar{y}_{fn} = \bar{\psi}_{fn}(\mu - \underline{c}_{fn})^{1-\sigma}, \quad \underline{y}_{gn} = \psi_{gn}(\mu + c_{gn})^{1-\sigma}, \quad \mu_- = \frac{\sigma}{\sigma - 1}, \quad \mu_+ = \frac{\eta}{\eta - 1}.$$

The markup interval gives

$$\bar{s}_{fn\varsigma} = \frac{\bar{y}_{fn}}{b_n + \bar{y}_{fn} + \sum_{g \neq f} \underline{y}_{gn}}. \quad (\text{B.1})$$

Inactive rivals are omitted and own share is zero if no completion supplies the market. The residual contribution b_n stays fixed. Complete AB portfolios are evaluated at joint equilibrium prices; the markup envelope is used only to bound completions.

Cardinality conditioning weakly improves the maximal-portfolio envelope when both use the same continuation envelopes: every top- k sum is at most the sum over U , and $\text{Bot}_k(C_U) \geq \sum_{j \in U} \min\{C_j, 0\}$. This does not imply dominance over bounds constructed from different envelopes. Independently valid caps can be combined by taking their minimum.

B.2 A common affine ranking of benefits and costs

An optimistic portfolio can combine benefits that no set of locations delivers together. The following bound charges for the missing benefits and ranks locations using those charges and their actual costs. Fix (S, U) and a completion size k . Index market-rival-state pairs by a , with weights $\omega_a \geq 0$. Let $G_a(t)$ be repriced market profit at own log competitiveness t , holding the rival state fixed, and let $c_j = C_{fj}$. All conditions below cover the relaxed family $\mathcal{H}_k = \{H \subseteq U : |H| = k\}$.

Proposition B.1 (Joint-affine completion bound). *Suppose positive endpoints $\bar{\xi}_a$ and affine coefficients satisfy*

$$\log \frac{\xi_a(H)}{\bar{\xi}_a} \leq d_a + \sum_{j \in H} A_{aj} \leq 0, \quad H \in \mathcal{H}_k. \quad (\text{B.2})$$

Suppose $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$ and $G'_a(t) \geq \mu_a \geq 0$ throughout every interval from $\log \xi_a(H)$ to $\log \bar{\xi}_a$. Define $\bar{G} = \sum_a \omega_a \bar{G}_a$, $D = \sum_a \omega_a \mu_a d_a$, and $\beta_j = \sum_a \omega_a \mu_a A_{aj}$. Then

$$B^{\text{joint}}(S, U, k) = \bar{G} - C(S) + D + \text{Top}_k(\beta - c) \quad (\text{B.3})$$

bounds every feasible completion and, using the same endpoint caps,

$$B^{\text{joint}}(S, U, k) \leq B^{\text{card}}(S, U, k) := \bar{G} - C(S) - \text{Bot}_k(c). \quad (\text{B.4})$$

For unrestricted size, maximize over admissible k .

Proof. Fix a completion H . Its shortfall from the optimistic log score is at least $g_a(H) = -d_a - \sum_{j \in H} A_{aj} \geq 0$. The slope lower bound converts this shortfall into a guaranteed loss of operating profit:

$$\begin{aligned} G_a(\log \bar{\xi}_a) - G_a(\log \xi_a(H)) &\geq \mu_a [\log \bar{\xi}_a - \log \xi_a(H)] \\ &\geq \mu_a g_a(H). \end{aligned} \quad (\text{B.5})$$

Write $C(H) = \sum_{j \in H} c_j$ and $L(H) = \sum_a \omega_a \mu_a g_a(H) = -D - \sum_{j \in H} \beta_j$. Thus $L(H)$ is a lower bound on the forgone operating profit, not the loss itself. Since $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$, expected net profit is at most $\bar{G} - C(S) - [C(H) + L(H)]$. Both the cost and the shortfall must come from the same locations. Minimizing their sum gives

$$B^{\text{joint}} = \bar{G} - C(S) - \min_{H \in \mathcal{H}_k} \{C(H) + L(H)\} = \bar{G} - C(S) + D + \text{Top}_k(\beta - c).$$

The gain over the cardinality bound is therefore

$$B^{\text{card}} - B^{\text{joint}} = \min_{H \in \mathcal{H}_k} \{C(H) - \text{Bot}_k(c) + L(H)\}.$$

Each term inside braces is nonnegative, proving dominance without a restriction on cost signs. $\square$

The improvement is strict exactly when no cheapest completion has zero shortfall $L(H)$. Strictness at one k need not survive maximization over sizes.

B.3 Service envelopes for the affine bound

Consider additive service and accessibility statistics

$$\xi_a(H) = \zeta_a \Sigma_a(H)^{v_S} W_a(H)^{v_A}, \quad \Sigma_a(H) = \Sigma_a^0 + \sum_{j \in H} v_{aj}, \quad W_a(H) = W_a^0 + \sum_{j \in H} u_{aj}, \quad (\text{B.6})$$

with nonnegative contributions and exponents, and positive statistics whenever their exponent is positive. Bases include selected and mandatory facilities; the accessibility base also includes the log-sum constant. In the cloud model $v_S = (\sigma - 1)/\theta$ and $v_A = \chi/\lambda_A$. Fixed normalizations enter $\zeta_a > 0$. Set $\bar{\Sigma}_a = \Sigma_a^0 + \text{Top}_k(v_{a\cdot})$, $\bar{W}_a = W_a^0 + \text{Top}_k(u_{a\cdot})$, and $\bar{\xi}_a = \zeta_a \bar{\Sigma}_a^{v_S} \bar{W}_a^{v_A}$. The inequality $\log z \leq z - 1$ supplies (B.2) with

$$d_a = v_S \frac{\Sigma_a^0 - \bar{\Sigma}_a}{\bar{\Sigma}_a} + v_A \frac{W_a^0 - \bar{W}_a}{\bar{W}_a}, \quad A_{aj} = \frac{v_S v_{aj}}{\bar{\Sigma}_a} + \frac{v_A u_{aj}}{\bar{W}_a}. \quad (\text{B.7})$$

Zero exponents remove the corresponding term before evaluating ratios. Inactive completions are handled separately without taking a log score.

Holding rival scores and residual contribution b fixed, Lemma A.3 supplies $\mu = \underline{\kappa}_{AB} R \min\{u(s^L), u(s^U)\}$ for a share interval covering every path from a completion to its synthetic upper endpoint. Let $\xi^L \leq \xi_f \leq \bar{\xi}$, $m_{\min} = (1 - 1/\eta)^{\sigma-1}$, and $m_{\max} = (1 - 1/\sigma)^{\sigma-1}$. The latter two values are the minimum and maximum of the markup term $\mu(s)^{1-\sigma}$, respectively, so

their order reverses the markup order. One such interval is

$$\begin{aligned} s^L &= \frac{\xi^L m_{\min}}{\xi^L m_{\min} + \sum_{g \neq f} \xi_g m_{\max} + b}, \\ s^U &= \frac{\bar{\xi} m_{\max}}{\bar{\xi} m_{\max} + \sum_{g \neq f} \xi_g m_{\min} + b}. \end{aligned} \tag{B.8}$$

Bottom- k sums give a lower score. A zero denominator can instead use $[0, 1]$ and $\mu = 0$. These are derivative bounds, not concavity assumptions.

Parent-dependent service. The actual mapping need not be additive. It suffices to construct additive statistics $\Sigma_a^+(H) \geq \Sigma_a(H)$ and $W_a^+(H) \geq W_a(H)$, including all parent-supported paths and permitting duplicate supply. Write $\xi_a^+(H) = \zeta_a[\Sigma_a^+(H)]^{v_S} [W_a^+(H)]^{v_A}$ and construct $\bar{\Sigma}_a, \bar{W}_a, d_a, A_{aj}$ from these additive statistics as above. For positive scores, monotonicity and $\log z \leq z - 1$ give

$$\log \frac{\xi_a(H)}{\bar{\xi}_a} \leq \log \frac{\xi_a^+(H)}{\bar{\xi}_a} \leq d_a + \sum_{j \in H} A_{aj} \leq 0.$$

The policy searches first dominate actual profit by profit at the relaxed score, and then apply the slope correction to that relaxed score. Since repriced profit is nondecreasing,

$$G_a(\log \xi_a(H)) \leq G_a(\log \xi_a^+(H)) \leq \bar{G}_a + \mu_a \left(d_a + \sum_{j \in H} A_{aj} \right).$$

The slope bound need only cover the relaxed score through the upper endpoint. Bottom- k sums of the additive contributions provide a lower bound for that relaxed score, not necessarily for actual supply. Summing and ranking the same location costs gives the stated joint-affine bound. Zero relaxed supply is handled separately; setting $\mu_a = 0$ retains the endpoint cap whenever no valid positive slope correction is available. Applying Proposition B.1 directly to actual scores is also valid if its slope bound covers their entire intervals to the upper endpoints.

The endpoint cap must cover profit at the synthetic score itself, even when that score is unattainable. A cap covering only feasible portfolios cannot safely receive the slope correction. Appendix B.5 discusses numerical allowances.

B.4 A bound retaining common core and service choices

Exact score reduction. Fix a firm, regime, and rival distribution, suppressing their indices. Under the change of variables in (3.8), $\xi_n = \zeta_n z_n^{v_S + v_A}$. In the calibration $v_S = 4/3$, $v_A = 4/15$, and $z_n = \Sigma_n^{5/6} W_n^{1/6}$, including zero supply. With positive residual supply, Lemma A.3 and continuity

at zero imply that repriced own profit is nondecreasing in ξ_n . Hence

$$H_n(z) = \sum_{\zeta} q_{\zeta} \pi_{fn}(\zeta_n z^{\nu_S + \nu_A}; S_{-f}^{\zeta})$$

is nondecreasing. The expression averages state-specific pricing equilibria, not rival supply statistics. It need not be concave.

Physical feasibility. Write x_i for core presence and o_i for coverage from either a core or a supported local path at origin i . For a local path h requiring all cores in a nonempty parent set P_h , introduce t_h and impose

$$0 \leq x_i, o_i, t_h \leq 1, \quad t_h \leq x_j \ (j \in P_h), \quad t_h \geq \sum_{j \in P_h} x_j - |P_h| + 1.$$

If $\mathcal{L}_i$ lists the alternative local paths at origin i and $\bar{o}_i \in \{0, 1\}$ indicates fixed local availability, impose $o_i \geq \bar{o}_i$, $o_i \geq x_i$, $o_i \geq t_h$ for $h \in \mathcal{L}_i$, and $o_i \leq \bar{o}_i + x_i + \sum_{h \in \mathcal{L}_i} t_h$. Together with $o_i \leq 1$, this includes fixed local supply without requiring an optional parent core. These constraints reproduce the core/local mapping at binary core choices; their continuous relaxation contains all physical completions. For geographic eligibility e_{in} ,

$$\begin{aligned} \Sigma_n(x, o) &= \Sigma_n^0 + \sum_i e_{in} (w_i \tau_{in})^{-\theta} [(1 - \gamma_{\text{loc}}) x_i + \gamma_{\text{loc}} o_i], \\ W_n(x, o) &= W_n^0 + \sum_i e_{in} e^{\lambda_A q_{in}} o_i. \end{aligned}$$

Ordinarily $\Sigma_n^0 = 0$ and $W_n^0 = 1$. China's fixed partner supply enters these constants only for eligible destinations, with its actual supply weight and overlapping optional paths removed. Forbidden direct cores have $x_i = 0$. Fix $x_i = 1$ on S_f , $x_i = 0$ outside $S_f \cup U_f$, and $\sum_{i \in U_f} x_i = k$ to obtain $\mathcal{R}_f(S_f, U_f, k)$. Other feasibility restrictions may be added provided every admissible completion remains represented. No claim that this relaxation is the convex hull is required.

Proof of Proposition 5. Every feasible completion has a representation in $\mathcal{R}_f$. Set z_n to its actual weighted geometric mean and $h_n = H_n(z_n)$. The upper-envelope property makes this point feasible in (3.9), and its objective equals the completion's expected profit net of all costs, including selected locations and any negative costs. This proves the bound. Convexity serves a separate purpose: it makes the relaxed bound tractable. The weighted geometric mean is concave on the nonnegative orthant, and $\hat{H}_n$ is concave. Their hypographs are convex; the remaining constraints and objective are linear (Boyd and Vandenberghe, 2004, Section 3.1.7). Maximizing over completion sizes therefore bounds unrestricted portfolio choice. $\square$

Constructing and evaluating the upper bound. On a grid $z_0 < \dots < z_L$ covering I_n , let $u_i \geq H_n(z_i)$ be nondecreasing upper values, replacing them by cumulative maxima if needed. Here H_n includes segment expenditure. Monotonicity gives $H_n(z) \leq u_{i+1}$ on $[z_i, z_{i+1}]$. The upper concave hull of (z_i, u_{i+1}) for $i < L$ and (z_L, u_L) is therefore a nondecreasing piecewise-linear concave majorant: both endpoints of each grid interval dominate its step bound. This controls values between grid points without assuming concavity of actual profit. Finer grids reduce discretization error but need not remove the gap from replacing a nonconcave function by a concave upper envelope.

With $\vartheta = \nu_S/(\nu_S + \nu_A)$, concavity of the weighted geometric mean gives, for every $\rho > 0$,

$$\Sigma^\vartheta W^{1-\vartheta} \leq \vartheta \rho^{\vartheta-1} \Sigma + (1-\vartheta) \rho^\vartheta W.$$

Replacing its hypograph by any finite collection of these affine upper constraints, and writing $\widehat{H}_n(z)$ as the minimum of its affine pieces, gives an LP relaxation containing every feasible completion. The LP may be looser than (3.9). If $\overline{B}_{f,k}^{\text{LP}}$ is a valid upper bound on its maximum and $B_{f,k}^{\text{old}}$ is an existing valid cap, then

$$B_f^{\text{used}} = \max_k \min\{B_{f,k}^{\text{old}}, \overline{B}_{f,k}^{\text{LP}}\} \leq \max_k B_{f,k}^{\text{old}}$$

is a valid completion bound. This comparison does not assert that the raw LP bound dominates the old cap. Numerical pruning requires an upper bound from a dual solution with controlled residuals; a feasible primal objective alone is insufficient.

Finally, if adding a given core never reduces operating profit and preserves feasibility for every portfolio omitting it, every unrestricted optimum contains that core when its realized cost is strictly negative: adding it strictly increases net profit. The maintained cloud mapping has this monotonicity. This dominance rule applies before searching across all sizes; it need not hold within a separately fixed-size problem. Neither this rule nor the geometric relaxation requires an ordering of marginal location returns.

B.5 Branch search and numerical error

For a fixed type and rival distribution, a node (S, U) represents the feasible portfolios T satisfying $S \subseteq T \subseteq S \cup U$. Start with a fully repriced feasible incumbent. Discard a node if infeasible or if its valid upper bound does not exceed that incumbent; evaluate a complete node; otherwise split on inclusion of an undecided country. The children partition the completion set without ordering marginal returns. Best-bound selection changes processing order, not validity. Let L be the best attained value and $\mathcal{O}$ the open nodes. If pruning uses $B(N) \leq L + \epsilon$, retain the largest bound U_{pruned} of those discarded nodes. At interruption,

$$U^* = \max\{L, \max_{N \in \mathcal{O}} B(N), U_{\text{pruned}}\}, \quad \max \emptyset = -\infty,$$

satisfies

$$L \leq \max_T v_f(T) \leq U^*.$$

The search ends to tolerance ϵ when $U^* - L \leq \epsilon$. With exact pruning, $U_{\text{pruned}} \leq L$ and that term is redundant. Finiteness ensures eventual termination, not a small gap within a time limit.

The searches underlying the policy tables apply bounds within each admissible size and retain the largest remaining bound across sizes. A route envelope includes every path supportable by a completion; an additive relaxation attributes local service to possible parents, permitting duplicates and relaxing joint parent requirements. Componentwise top- k sums bound that relaxation. The smaller competitiveness envelope is repriced against the supplied rival states and combined with the corresponding cost lower bound and joint-affine cap. The EU search permits every subset of 42 core countries. China excludes direct mainland core choice for the five non-Chinese providers; fixed AWS and Microsoft partner paths enter exact payoffs and all upper envelopes identically, without a selectable core or private core cost. The geometric cap is evaluated separately in the controlled speed comparison of Appendix C.2; that comparison does not replace the audits of the reported policy tables.

Numerical and distributional error. If the active log-contribution vector $x = (\log y_i)_i$ has undamped pricing residual r_x , the conditional contraction bounds its sup-norm error by $r_x/(1 - \bar{\lambda}_{AB})$. Residuals in shares must first be converted to these units. Profit allowances must enlarge node bounds and lower incumbents; floating-point residuals alone do not bound rounding error. Shares, slopes, affine coefficients, and sums must be enclosed, including cancellation and signed costs. Taking the minimum with a valid cap cannot repair an invalid one. Complete portfolios are repriced using the original payoff, rather than the upper envelope.

Bounds built at finite rival quadrature apply to that payoff; the full expectation needs an additional integration allowance. At a specified cost type and rival distribution, a full-choice-set search bounds all feasible portfolio deviations. It does not control integration over other cost types or best-response error after rivals change. Discovered portfolios enter common menus before policy distributions are recomputed and searched again; independent cost draws test coverage beyond previously searched types.

Appendix F separates these checks. Differences across quasi-Monte Carlo scrambles measure numerical stability, not sampling confidence intervals. A uniform contraction would additionally convert a response residual to a distributional distance through Proposition A.1. Without it, regret, total variation, and agreement across starts remain distinct diagnostics; small profit errors need not imply small policy-price errors.

C. METHOD DIAGNOSTICS

C.1 A unique portfolio equilibrium despite sign reversals in location returns

Consider two firms, one market, three optional locations per firm, unit expenditure, residual contribution 0.005, and $(\sigma, \eta, \theta) = (5, 2, 3)$. Firm f has unit accessibility and cost $x_f(S)^{-1/3}$, where $x_f(S) = x_{0f} + \sum_{j \in S} d_{fj}$. All subsets are feasible. Base supplies are $(.001, 1)$, optional contributions are $(.02, .02, .099)$ and $(.1, .1, .2)$, and deterministic costs are $(.0064, .0064, .0283)$ and $(.0063, .0063, .0125)$. Each optional location adds independent cost $0.01\varepsilon_{fj}$, with standard normal shocks, so $\sigma_\varepsilon = 0.01$ and $\ell_{fj} = 1$. At zero own shocks and an empty rival optional portfolio, firm 1's net returns to location 1 along $\emptyset \subset \{2\} \subset \{2, 3\}$ are approximately -8.51×10^{-4} , $+5.47 \times 10^{-4}$, and -6.03×10^{-4} . These strict signs persist on a neighborhood of own costs and violate both directions of SCD-C (Definition 3 of Arkolakis et al. 2026), as well as submodularity and supermodularity.

Evaluating all 64 joint portfolios and all 28 unordered rival pairs per firm gives the normal-cost upper matrix

$$\bar{K}^N = \begin{pmatrix} 0 & 0.003741736 \\ 0.004367158 & 0 \end{pmatrix}, \quad \rho(\bar{K}^N) < 0.004043 < 0.01.$$

Thus the continuous-type portfolio equilibrium is unique despite nonordered location returns. The entries of $\bar{K}^N$ round upward from payoff enclosures obtained using the pricing contraction and interval arithmetic; the primitive decimals are exact inputs. These constructed primitives establish coexistence of the two properties, not typical performance in the calibration.

C.2 Comparison with exhaustive enumeration

The experiment fixes the EU baseline rival distribution and uses all eight providers, two independent Gaussian cost vectors per provider, and the applications' 1,024-point rival integration rule. A prespecified random permutation gives nested sets of 8, 10, 12, 14, 16, and 20 optional core countries, common to all firms. Decisions outside each set are fixed to the observed portfolio; portfolio size within the set is endogenous. All 42 customer countries and the core-local service mapping remain. The country sequence and cost vectors are selected before inspecting computation times or profits.

Three methods solve each problem. Enumeration prices every feasible portfolio using the same compiled pricing routine as the search methods. One search uses the cardinality and joint-affine bounds. The strengthened search also uses the common-location LP bound, cardinality-dependent supply intervals, and reusable constraint matrices. It evaluates the LP before calling the cardinality and affine bounds when further tightening is needed. Cost-dependent presolve is disabled in all

Table 8: Computation time by number of optional locations

Optional locations	Median time (seconds)			Median time ratio	
	Enumeration	Cardinality and affine	With common locations	Enumeration/ common locations	Cardinality–affine/ common locations
8	4.5	5.9	18.6	0.24	0.32
10	12.6	12.3	20.4	0.62	0.58
12	45.0	30.7	22.6	1.99	1.28
14	173.6	57.2	25.1	6.93	2.14
16	700.8	125.6	28.5	24.47	3.92
20	11,956.8	631.6	50.9	235.33	10.57

Notes: Each row covers all eight firms and two cost vectors per firm. Times include preparation and solving both types. Ratios are medians of firm-specific time ratios, not ratios of the time medians. Two firms run concurrently at 8–16 locations and eight at 20 locations; methods within a firm run sequentially in rotating order, with one computation thread. These are single-run elapsed times on the same Apple M4 Pro computer, subject to host-load variation. “With common locations” adds the common-location LP bound to the cardinality and joint-affine bounds. Both search methods are developed in this paper.

three methods to hold the number of optional locations fixed. Initial portfolios are the same observed and cost-ordered candidates, repriced separately by each method; enumeration solutions are never passed to the searches.

Each method runs in a fresh process and may reuse its own operating profits across the two cost vectors. Timing starts before input loading, initial candidate evaluation, and profit-curve construction. Independent recalculation of prices for the selected portfolios follows the timed portion for every method. The three methods agree on optimized profits in all 96 type problems, and every search meets the common profit-gap tolerance. At 20 optional locations, each firm’s enumeration evaluates all 1,048,576 portfolios; none of the reported times is extrapolated from an unfinished run.

These restricted problems measure global best-response computation on one prespecified sequence, not equilibrium iteration, unrestricted 42-country performance, or comparisons with methods requiring ordered returns.

C.3 Portfolio choices under alternative pricing rules

A separate eight-provider exercise compares globally optimal choices under AB and constant-markup CES pricing in 24 firm–regime problems, with 256 private-cost types per problem. Core presence can adjust in ten countries; rival portfolios and own choices outside those countries are fixed at their observed values. Optimal AB and CES choices differ for 80.66, 79.54, and 77.93 percent of types in the baseline, EU-wide, and National regimes, respectively. Both specifications use the same AB-calibrated parameters; CES is not recalibrated. This is a comparison of pricing assumptions within a restricted choice set, not a comparison of empirical fit or solution accuracy.

Choice disagreement does not measure the error in a policy counterfactual, and the exercise does not re-solve portfolio equilibria.

C.4 Failure of the global sufficient condition for uniqueness under normal private costs

The midpoint formula (A.9) implies $C_f^N(h) \geq |\Delta_j h(S)|/(\ell_{fj}\sqrt{2\pi})$ for every adjacent portfolio pair: its norm term bounds $|b_j|/\ell_{fj}$, its remaining term bounds r_j/ℓ_{fj} , and $|\Delta_j h(S)| \leq |b_j| + r_j$. In the calibrated EU baseline, compare adding a US core to an otherwise empty own core network when the rival network is empty or spans all 42 countries. Keep the other six core networks empty and retain the local service and parent-path rules. AWS's operating-profit increment changes from 0.142787 to 0.051828; Microsoft's changes from 0.043477 to 0.008863. Unrounded payoff enclosures and the calibrated private-cost loadings give the downward-rounded bounds

$$K_{\text{AWS,MSFT}}^N \geq 1.523058, \quad K_{\text{MSFT,AWS}}^N \geq 1.240307, \quad \frac{\rho(K^N)}{0.5} > 2.74.$$

Deterministic additive costs cancel. These feasible configurations lower-bound the global coefficient regardless of their equilibrium probability. Refining an upper bound on the same coefficient cannot make its sufficient test pass. This is neither evidence of multiplicity nor a test of the localized or signed-market conditions.

D. SERVICE AGGREGATION, DATA, AND CALIBRATION UNITS

We derive the unit-cost technology and then describe the data conventions. The matching equations and parameter values are in Section 4.2.

D.1 Fixed workload composition and unit cost

Fix a provider, segment, regime, and portfolio, suppressing their indices. A unit of service requires workloads $v \in [0, 1]$ in fixed proportions, represented by a measure of mass one. Eligible origins and intensities t_i are common across these workloads. For $t_i > 0$, efficiency has distribution $\Pr\{Z_i(v) \leq z\} = \exp(-t_i z^{-\theta})$, with $\theta > 0$ and independence across origins. Each workload is performed at its least-cost eligible origin. Writing $\tilde{a}$ for the common efficiency scale,

$$m(v) = \min_{i:t_i>0} \frac{w_i \tau_{in}}{\tilde{a} Z_i(v)}, \quad \Pr\{m(v) > x\} = \exp(-\tilde{a}^\theta \Sigma x^\theta), \quad \Sigma = \sum_i t_i (w_i \tau_{in})^{-\theta}. \quad (\text{D.1})$$

Assume the workload aggregate equals the distributional integral. For $\Sigma > 0$, fixed requirements imply a unit cost

$$c = \int_0^1 m(v) dv = \int_0^\infty e^{-\tilde{a}^\theta \Sigma x^\theta} dx = \frac{\Gamma(1 + 1/\theta)}{\tilde{a}} \Sigma^{-1/\theta}. \quad (\text{D.2})$$

Substitution of $u = \tilde{a}^\theta \Sigma x^\theta$ gives the last equality, finite for every $\theta > 0$. Defining $a = \tilde{a}/\Gamma(1 + 1/\theta)$ gives (2.1); the calibrated a_f is already in these units. With no eligible origin, supply is unavailable. This integrated workload heterogeneity is distinct from the private location-cost shocks.

For the core/local rule, local opportunities have intensity γ_{loc} ; a core contains these and independent additional opportunities of intensity $1 - \gamma_{\text{loc}}$. Their maximum has intensity one, preserving local supply when a core is added without double counting. This nesting is an assumption about compatible workloads, not observed product availability.

An internal CES aggregator with elasticity $\rho > 0$, $\rho \neq 1$, would use the moment $\mathbb{E}m^{1-\rho}$, requiring $\theta > \max\{\rho - 1, 0\}$; the Cobb–Douglas limit at $\rho = 1$ is also finite for $\theta > 0$. Antràs et al. (2017, eqs. (5) and (9)) separate internal aggregation from external demand. Setting $\rho = \sigma$ would impose $\theta > \sigma - 1$, as in Tintelnot (2017). Fixed workload composition, the $\rho = 0$ limit derived above, permits $(\sigma, \theta) = (5, 3)$.

Workload classes with different eligibility sets require separate supply aggregates: fixed class quantities imply a sum of class costs, not a single aggregate with intensity equal to the fraction of supported products. An unavailable required class makes the bundle infeasible. The applications use one representative class; public shares do not identify class weights.

D.2 Expenditure, delivery, and accessibility inputs

Country expenditure is the dimensionless proxy

$$\bar{R}_n = \left[\frac{\text{GDP}_{n,2023}(I_{n,2023}/100)}{\text{GDP}_{\text{US},2023}(I_{\text{US},2023}/100)} \right]^{0.7}, \quad \bar{R}_{\text{US}} = 1.$$

GDP is nominal GDP in current U.S. dollars; I is the percentage of the population using the internet, not the number of users. The inputs use 2023 World Bank indicators, with an IMF GDP estimate and a maintained 93 percent internet-use rate for Taiwan. The exponent 0.7 compresses cross-country differences and is held fixed. The proxy is not observed cloud revenue. For EU countries the customer segments have expenditure $R_{n,\text{reg}} = 0.18\bar{R}_n$ and $R_{n,\text{private}} = 0.82\bar{R}_n$; other countries retain $\bar{R}_n$. These expenditure weights are unchanged across policy regimes.

The baseline delivery wedges take the values in (4.1). The starting accessibility score is

$$q_{in}^0 = -0.25 \log(1 + d_{in}/1000) + 0.201\{i = n\} + 0.151\{\tau_{in} \leq 1.15\} + 0.20(\omega_n - 1), \quad (\text{D.3})$$

where d_{in} is distance in kilometers between the input country's reference coordinates. These include facility-based geographic averages and city fallbacks. The final term is a destination-specific

index from the 2025 OECD digital-policy inputs, held fixed across counterfactuals, with neutral value $\omega_n = 1$ when unavailable. Its coefficient is assumed; the index is neither a provider revenue share nor a measured willingness to pay. The proximity indicator uses the baseline delivery wedge, so a counterfactual change in τ does not mechanically change q .

For each EU destination, we reorder the eleven foreign-EU scores in (D.3) using RIPE round-trip-time rankings. The country-pair statistic is the median of anchor-pair minimum ping measurements from the August 18, 2026 archive. The sample retains anchor pairs observed in both directions, while computing each direction's statistic separately. Shorter round-trip times receive higher scores. Domestic scores, non-EU destinations, and non-EU origins retain q^0 . This replaces the assignment of scores within 132 directional EU country pairs; it does not estimate a latency coefficient or use latency differences elsewhere to change supply costs.

Reordering preserves the sum of $e^{\lambda_A q_{in}}$ over origins at each destination. We use the fixed all-origin frontier

$$A_n^{\text{frontier}} = \lambda_A^{-1} \log \left(1 + \sum_{i=1}^{42} e^{\lambda_A q_{in}} \right),$$

computed before applying policy eligibility. Both the ranking replacement and later policy comparisons therefore leave this reference unchanged. Otherwise an eligibility restriction could lower a provider's access score and its normalization simultaneously, obscuring the service loss. This convention does not imply that any provider operates the frontier network.

The central residual term is common across markets: $b_n = b^0 \exp(\zeta_b)$, with $b^0 = 1.77817$ and calibrated log multiplier $\zeta_b = 0.1396$ (rounded). EU public and private segments inherit the same term. Its realized share is endogenous because the contributions of the eight providers change. Residual sensitivities vary this term or its exposure to the China shock without treating it as a separate strategic firm. The common profile is a maintained simplification of unmodeled supply, not a fitted map of regional supplier capacity.

D.3 Profit and cost scales

The reference cardinalities used to normalize private costs and gains in Table 9 differ from the current observed core counts, especially for Alibaba, Huawei, Tencent, and OVHcloud. For example, OVHcloud's reference count is 18, whereas its observed core-country count is 10. These reference-network quantities were set before the current endogenous-count calibration and preserve a fixed provider-specific cost scale across its specifications. They are not observed monetary costs; using observed core counts instead would define a different scale assumption. With the other coefficients fixed, changing $\ell_f^{\text{ref}} = G_f^{\text{ref}} / K_f^{\text{ref}}$ changes both deterministic location costs and private-cost dispersion. The loadings are maintained scale assumptions, held fixed as portfolios and policy regimes change.

Table 9: Profit and cost scales

Provider	ℓ_f^{ref}	K_f^{ref}	Observed cores	G_f^{ref}
AWS	0.02383	28	28	0.66711
Microsoft	0.01113	33	33	0.36740
Google	0.00398	29	29	0.11553
Oracle	0.00178	27	27	0.04796
Alibaba	0.00604	15	17	0.09054
Huawei	0.00494	10	12	0.04938
Tencent	0.00212	11	13	0.02332
OVHcloud	0.00006	18	10	0.00113

$G_f^{\text{ref}} = \ell_f^{\text{ref}} K_f^{\text{ref}}$. These inherited constants scale private costs and payoff gains and are held fixed across calculations. Reference counts differ from current observed counts. Changing the loadings with the other coefficients fixed changes cost levels and dispersion, rather than merely changing units. Unrounded values are included in the replication data.

The missing-construction-data penalty $\delta = 0.614$ was inherited from a location-composition fit that held provider core counts fixed. It was retained when the construction-index slope was removed. It is not re-estimated in the present count-and-share calibration and is not an externally measured premium for these countries. It applies to Belgium, Bahrain, Hong Kong, Israel, Morocco, the Philippines, Qatar, Serbia, Thailand, and Taiwan. Belgium is the only model-EU country in this list. Thus the penalty is not an EU-wide entry-cost adjustment; it also changes the attractiveness of non-EU alternatives. The fixed-network price comparisons do not assess how changing this cost term would alter portfolio choices.

D.4 Geographic composition diagnostics

Forty correlated composition moments informed specification assessment: for each provider, the fractions of core presences in the model EU, Asia-Pacific, and the Americas, the mean median-normalized construction-cost index, and mean log expenditure in hosting countries. The reported calibration does not fit additional spatial coefficients to these moments. Missing construction costs use a projection on public covariates and a median fallback. With variable portfolio size, the moment for country feature h_{ij} is

$$m_{fj}(Q_f) = \frac{\mathbb{E}_{Q_f}[\sum_{i \in S_f} h_{ij}]}{\mathbb{E}_{Q_f}[|S_f|]}. \quad (\text{D.4})$$

The ratio of expectations weights countries by their inclusion probabilities. It differs from averaging each portfolio's fraction and retains zero-core types in the unconditional expectations. Composition is undefined if the expected core count is zero. Table 10 shows that the model compresses regional specialization: across providers, its EU share of core locations lies between 26 and 33 per-

cent, against 8 to 40 percent observed. The excess EU presences come mainly from AWS, Oracle, and the three Chinese providers, while OVHcloud’s EU share is underpredicted.

Table 10: Geographic composition: observations and model

Provider	EU (%)		Asia-Pacific (%)		Americas (%)	
	Obs.	Model	Obs.	Model	Obs.	Model
AWS	21.4	32.5	42.9	26.0	14.3	17.8
Microsoft	33.3	32.0	30.3	27.7	15.2	16.5
Google	31.0	32.9	31.0	26.5	17.2	17.8
Oracle	22.2	33.1	25.9	26.1	22.2	18.4
Alibaba	11.8	28.0	52.9	45.2	17.6	8.7
Huawei	8.3	25.9	50.0	50.3	25.0	7.5
Tencent	7.7	26.7	61.5	46.9	23.1	8.6
OVHcloud	40.0	28.4	30.0	21.2	20.0	11.6

Provider	Construction index		Log expenditure	
	Obs.	Model	Obs.	Model
AWS	-0.002	-0.002	-2.185	-2.269
Microsoft	0.005	-0.009	-2.232	-2.350
Google	0.009	-0.013	-2.203	-2.322
Oracle	0.003	-0.014	-2.249	-2.313
Alibaba	0.013	0.000	-1.961	-2.333
Huawei	-0.038	-0.015	-2.420	-2.250
Tencent	0.022	-0.011	-1.889	-2.284
OVHcloud	-0.028	-0.032	-1.676	-2.474

Notes: Each cell reports the ratio in equation (D.4). Observations use core-country presences; model values use the same baseline portfolio distribution as Table 3. Construction costs enter as their ratio to the median observed cost minus one, with ICP-based imputation and a median fallback for missing observations. Expenditure is the GDP–internet proxy normalized to the US. The three regions do not exhaust the country universe. These forty correlated diagnostics informed specification assessment; they are neither additional matching targets nor an unused validation sample.

E. CLOUD-NETWORK MEASUREMENT AND INSTITUTIONAL SOURCES

This appendix documents the current core-country and service inputs, parent connections, workload scope, and regulatory qualifications used to interpret the model.

E.1 Constructing core and service availability

Multiple named core regions within a country form one selectable core presence. Installed local facilities contribute service but do not count as additional cores. Confirmed parent support is evaluated for each complete portfolio before averaging; unknown parents retain unmodeled support, with no parent country imputed. Core-composition moments and service coverage therefore use distinct country counts. In the China experiment, fixed partner paths supply service and branded profit but carry neither selectable core counts nor private core-cost shocks.

The local-service ledger starts from 82 source records; 60 pass the 42-country and product screens. In the current inputs, 25 have a confirmed parent in the same country, nine have a confirmed parent abroad, and 26 have an unknown parent. The 34 confirmed assignments comprise 33 AWS records and Microsoft’s Los Angeles Extended Zone, linked to West US in its quota documentation. The unknown assignments comprise 25 OVHcloud records and Alibaba’s Wuhan Local Region. These are documentation counts, not provider-wide or capacity-weighted frequencies. Unknown parentage is not evidence that a local service is independent.

AWS’s architecture documentation, archived September 11, 2026, describes a Local Zone as an extension of a parent region but also documents direct internet and Direct Connect paths that need not traverse that parent. The parent assignments record documented product relationships rather than measured traffic routes.

E.2 Product scope and the representative workload

The current ledger records 169 core-country and 184 service-country presences across providers. Local paths add five countries for AWS and ten for OVHcloud; several facilities in one country remain one service origin. Neither a core nor a local record establishes availability of every product.

Table 11 distinguishes compute supply from product-specific restrictions. An unsupported accelerator or required storage feature makes a workload infeasible. Missing managed database services do not by themselves rule out running database software on a local virtual machine.

Table 11: Local supply and workload scope

Notes: The 60 local records are from the screened input ledger. Links identify official provider documentation archived in September 2026. Product lists establish availability within their documented scope, not relative capacity or the value of γ_{loc} .

Provider	Compute supply and product scope	Evidence boundary
AWS (33)	Virtual machines and block storage; some locations also support accelerators.	Zone-level feature lists differ across locations.
Microsoft (1)	Los Angeles virtual-machine deployment; selected compute, storage, and networking services.	Extended Zones documentation keeps the control plane in the parent region.
Alibaba (1)	Wuhan Local Region appears in the ECS roster.	Detailed product availability and parent support remain incompletely documented.
OVHcloud (25)	Virtual-machine supply; the capability guide excludes GPUs and limits some storage and network features.	Twelve records use a separate U.S. account; the guide is not a location-by-location verification of their product range.

The model’s representative workload captures the common compute margin; homogeneous eligibility and local intensity are maintained approximations. Worldwide IaaS revenue also includes GPU-oriented supply (Gartner, 2025), whose weight the public shares do not identify. Service counts do not measure workload shares or intensities. Product-specific eligibility would require the separate aggregates described in Appendix D.1.

E.3 Regulatory qualifications and operating modes

Table 12 separates customer eligibility from physical presence and operating arrangements. The underlying archive retains regulator documents, provider pages, retrieval dates, and page hashes. These records document supply arrangements; they do not identify rival responses, certification costs, or cross-market price effects.

Public-sector qualifications in France and Korea attach to particular services and operating arrangements rather than all infrastructure of a provider. US defense contracts provide a geographic example: DFARS 252.239-7010(b)(3) generally requires covered government data outside Department of Defense premises to remain in the United States or its outlying areas, subject to written exceptions (US Department of Defense, 2023). USMCA separates information transfers from facility-location requirements and excludes government procurement from its digital-trade chapter (Office of the United States Trade Representative, 2020, Articles 19.2, 19.11–19.12). USTR

Table 12: Regulatory access and cloud supply arrangements

Notes: The table reports documented institutional and supply distinctions. It does not assign a causal effect to a regulation. Qualification dates refer to the specified offer, not to a provider’s first country entry. “Held fixed” means that the feature is not an endogenous choice in the current quantitative model. Sources are listed below; Section E discusses their interpretation.

Case and covered demand	Supply boundary	Documented arrangement	Implication for the model
France: specified sensitive public-sector workloads	SecNumCloud qualification attaches to an offer and legal entity. Version 3.2 includes EU location and provider-control requirements.	S3NS, which uses Google Cloud technology under a Thales-controlled entity, qualified on December 17, 2025. Bleu uses Microsoft technology and remained in the qualification process in the September 2026 record. OVHcloud qualifies named offers.	Motivates buyer-specific eligibility. The EU experiment varies geographic recognition conditional on meeting the non-geographic qualification requirements; ownership and joint-venture choice are held fixed.
Mainland China: customers of locally supplied cloud services	Licensed local operators and counterparties supply the China regions in a separate contracting and account environment.	Sinnet and NWCD operate the two AWS China regions; 21Vianet operates Azure China. AWS documented a change in the Beijing-region counterparty on August 1, 2016 while referring generally to Chinese telecommunications regulation.	The baseline distinguishes brand, operator, and directly controlled core presence. The counterfactual changes cross-border costs relative to this segmented baseline.
Korea: public-sector workloads covered by the CSAP tier	Specified services require certification to serve public buyers, separately from the opening of a commercial region.	Microsoft certified services in Korea Central and Korea South on November 20, 2024; AWS certified services using its Seoul region and edge location on March 26, 2025. Google’s page confirms covered Seoul services but gives no certification date.	Motivates customer-specific eligibility distinct from core-country entry. Internal adaptation and certification costs are not observed and are held fixed.

Primary records: ANSSI SecNumCloud v3.2 and the ANSSI qualification catalogue; AWS–Sinnet operating relationship and Microsoft–21Vianet; Korean notice 2023-4, Microsoft CSAP, AWS CSAP, and Google Cloud CSAP.

discusses cloud rules as market-access concerns for US suppliers (Office of the United States Trade Representative, 2026).

F. POLICY ACCOUNTING AND NUMERICAL CHECKS

F.1 Candidate-set convergence and unrestricted search

The reported EU results use common candidate sets containing 9,850–12,284 portfolios per firm and 1,536 own-cost types, with calibrated parameters held fixed. Discovered portfolios enter every regime’s menu before its distribution is recomputed. Conditional AB prices are unique, with contraction modulus 0.375. Portfolio convergence is a separate question. The stopping rule, common to EU and China, requires each firm’s mean candidate-set regret to be at most $10^{-4}G_f^{\text{ref}}$ and

$$\max_f \frac{1}{2} \sum_{S_f} |\widehat{\mathcal{B}}_f(Q_{-f})(S_f) - Q_f(S_f)| \leq 0.01$$

at two consecutive checks. Regret includes the pricing allowance; the distributional residual is undamped and compares whole portfolios.

Table 13: Convergence and full-portfolio search diagnostics

<i>Panel A. Convergence on the candidate sets</i>				
	Maximum firm mean regret / G_f^{ref}	Maximum sampled type regret / G_f^{ref}	Undamped portfolio TV	
Baseline	2.80×10^{-6}	2.16×10^{-4}	0.00731	
EU-wide	3.38×10^{-6}	4.22×10^{-4}	0.00605	
National	6.50×10^{-6}	3.54×10^{-3}	0.00584	
<i>Panel B. Search over all feasible country portfolios</i>				
Regime	Problems	Improving portfolios found	Maximum remaining optimality gap / G_f^{ref}	Searches closed
Baseline	32	0	1.070	1
EU-wide	32	0	1.071	0
National	32	0	0.989	0

Notes: Panel A reports candidate-set diagnostics for simulated cost types; all three regimes meet the prespecified stopping rules. Panel B reports 96 full-choice-set search problems evaluated at four cost types per firm. No improvement beyond numerical error is found, but time limits leave most search bounds open. A remaining gap is the profit upper bound minus the best feasible profit found. The cost types are reused across search rounds. These checks do not prove optimality for every type in the continuous distribution or uniqueness of the portfolio equilibrium.

All three EU distributions satisfy the candidate-set stopping criteria. Full-choice-set searches find no additional improvement at four cost types per firm, reused across search rounds, but most remaining optimality gaps do not close (Table 13). The checks assess selected types at the sup-

plied rival distributions, rather than optimality throughout the continuous cost population. They concern the policy profiles, not the later speed comparisons in Appendix C.2.

All four China distributions also meet the candidate-set stopping rules. An easing search added one Alibaba portfolio; after recomputation on the common menu, the repeat search found no further improvement at four types per firm. Most global bounds remained open, with the largest gap about one reference payoff. Menu expansion changed central price and worldwide-count effects by less than 0.03 percentage points and 0.03 presences, respectively. Two outcome-integration samples differ by less than 0.01 percentage points in the central China price comparisons. The following subsection examines initialization and integration sensitivity.

F.2 Initialization and integration sensitivity

Each China regime is restarted from minimum-count, maximum-count, and uniform candidate distributions. Eleven of twelve runs meet the stopping rules; the remaining baseline run finishes just above the TV criterion and stays in all comparisons. Relative to the reported distributions, the largest country-inclusion difference is 0.26 percentage points and the largest worldwide count difference is 0.11. Price-effect signs are unchanged, with maximum departures of 0.08 percentage points in China and 0.04 outside. This tests selected starts, not all equilibria.

At the original integration resolution, fresh rival samples yield small regret but all twelve restarts and four reference distributions fail the TV criterion. Nearly tied portfolios can cause this pattern, although the check does not identify the cause. Re-solving all three EU and four China distributions with 4,096 rival integration points, using the same own-cost sample and menus, yields seven distributions that satisfy the stopping criteria; an independent rival-integration check also passes. Relative to the originally computed portfolio distributions on matched outcome samples, worldwide policy-count differences change by less than 0.10 and regulated price effects by less than 0.02 percentage points.

Own-cost sampling remains a separate limitation. Applying frozen decision rules to 1,536 fresh cost vectors per firm produces worldwide counts about five to eleven below those on the calibration sample. This holds rivals' portfolio distributions and expected profits fixed rather than solving new equilibria. The reported counts and responses therefore retain the calibration sample; small rival-integration regret does not establish their stability across own-cost samples. Nor does the level discrepancy identify error in paired policy effects or competitor-response comparisons: those require common independent own-cost samples and re-solved distributions. Finally, stable policy statistics and unique conditional prices do not establish portfolio uniqueness or the containment required by the localized conditions.

Table 14: EU eligibility and residual supply across response comparisons

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
Eligible modeled providers	-2.48	-2.68	-2.62
Residual share (pp)	55.95	56.39	56.43

Notes: National minus EU-wide. Eligibility counts and shares are expenditure-weighted over regulated markets after profile-level pricing. Residual supply remains locally eligible.

F.3 EU accounting and sensitivity

Table 14 reports eligible-provider counts and residual shares after profile-level pricing. Table 6 separates residual intensity from eligibility: $b_n^N = (1-\phi)b_n^E$ makes fraction ϕ unavailable only under National. The eligibility comparison reprices fixed baseline portfolios with greater integration accuracy; portfolios are not reoptimized. The observed-core comparison substitutes the recorded cores while retaining the same installed local topology and parameters, measuring sensitivity to the entire starting geography.

One-firm responses. The distribution $Q^{r,\{f\}} = \hat{\mathcal{B}}_f^r(Q_{-f}^O) \otimes Q_{-f}^O$ allows only firm f to respond. Combining all responses to unchanged rivals instead gives $Q^{r,u} = \bigotimes_f \hat{\mathcal{B}}_f^r(Q_{-f}^O)$, which requires joint repricing. Its difference from the equilibrium comparison is not the sum of one-firm price effects. Hats denote common-menu approximations; the unrestricted searches target the recomputed policy distributions, not each unchanged-rival response. OVHcloud accounts for 42 percent of the equilibrium worldwide count reduction but counts do not measure price incidence. In the one-firm comparison it removes 2.305 worldwide core presences and raises the non-EU price index by only 0.002 percent. Tencent removes 1.382 and raises that index by 0.082 percent. The index weights service contributions, remaining alternatives, and provider shares, whereas counts weight every presence equally.

F.4 China accounting and residual exposure

China shares are predictions, not separate calibration targets. The model assigns AWS a larger mainland share than Alibaba, and Microsoft a comparable share. For context, Omdia (2026) reports 2025Q3 shares of 36 percent for Alibaba, 16 percent for Huawei, and 9 percent for Tencent. That market also includes PaaS and other infrastructure services, so these figures are not directly comparable workload-level targets. Nevertheless, the model’s supplier composition differs substantially from that evidence. Global competitiveness, country-level supply aggregation, offshore access, fixed partner paths, and residual supply jointly determine its China shares; the comparison does not identify which feature accounts for the mismatch. The five non-Chinese providers

Table 15: EU policy: provider responses and prices outside the EU

Responding provider	Worldwide core presences	Non-EU price (%)
AWS	-0.182	0.037
Microsoft	-0.174	0.038
Google	-0.370	0.041
Oracle	-0.833	0.047
Alibaba	-0.351	0.050
Huawei	-0.602	0.075
Tencent	-1.382	0.082
OVHcloud	-2.305	0.002
Responses to unchanged rivals	-6.199	0.378
Equilibrium responses	-4.112	0.243

Notes: National minus EU-wide. In each provider row, only that provider adjusts its portfolio, taking rivals' baseline portfolios as given; all firms' prices are re-solved. Responses to unchanged rivals combine the eight individual responses and jointly reprice them. Equilibrium responses use the policy-specific portfolio distributions. Core changes are expected provider-country counts. Price effects are nonlinear and cannot be added across provider rows.

Table 16: China provider shares and residual-response comparisons

Provider / response comparison	Baseline (%)	-9.52% (pp)	+10.52% (pp)	+28.40% (pp)
AWS	29.44	+0.62	-0.84	-2.47
Microsoft	17.50	+0.45	-0.63	-1.84
Google	5.23	+0.35	-0.43	-1.21
Oracle	2.21	+0.17	-0.19	-0.52
Alibaba	17.90	+0.59	-0.63	-2.02
Huawei	11.36	+0.31	-0.50	-1.37
Tencent	6.57	+0.29	-0.35	-1.17
OVHcloud	0.03	0.00	0.00	-0.01
Residual: equilibrium responses	9.78	-2.78	+3.59	+10.61
Residual: fixed networks	9.78	-2.72	+3.46	+10.02
Residual: unchanged-rival responses	9.78	-2.80	+3.69	+10.94

Notes: All comparisons use the same baseline and unexposed residual contribution ($\phi = 0$). Provider rows use equilibrium portfolio responses. The final two rows give alternative residual-share changes and are not part of the adding-up totals above. Shares are model predictions, with residual supply included in the denominator; AWS and Microsoft include branded partner supply. Rounding may prevent exact adding-up.

have zero direct-core probability by construction. AWS and Microsoft's fixed partner paths supply the mainland and earn branded profit without a selectable core or private core-cost draw. Partner

Table 17: China residual exposure at policy and baseline portfolios

Trade cost	China price (%)		Chinese provider share (pp)	
	$\phi = 0$	$\phi = 1$	$\phi = 0$	$\phi = 1$
<i>A. Each policy's central equilibrium portfolios</i>				
-9.52%	-8.05	-9.04	+1.19	-0.03
+10.52%	+8.17	+9.73	-1.49	+0.09
+28.40%	+20.25	+25.63	-4.56	+0.09
<i>B. Common baseline portfolios</i>				
-9.52%	-7.84	-8.84	+0.81	-0.41
+10.52%	+7.89	+9.43	-1.06	+0.52
+28.40%	+19.37	+24.53	-3.13	+1.56

Notes: Effects relative to the common baseline. At $\phi = 1$ all residual price-index contribution faces the offshore friction; all firms reprice but portfolios are not reoptimized. Panel A holds the central policy portfolios computed with $\phi = 0$; Panel B holds baseline portfolios throughout. Chinese providers are Alibaba, Huawei, and Tencent, measured as a share of the entire China market, including residual supply.

paths are excluded from worldwide core counts, so these counts are not mechanically comparable to EU counts or observed facilities.

Let fraction ϕ of residual contribution face the offshore cost change:

$$b_{\text{CHN}}(d; \phi) = b_{\text{CHN}}(0)[(1 - \phi) + \phi e^{-(\sigma-1)d}]. \quad (\text{F.1})$$

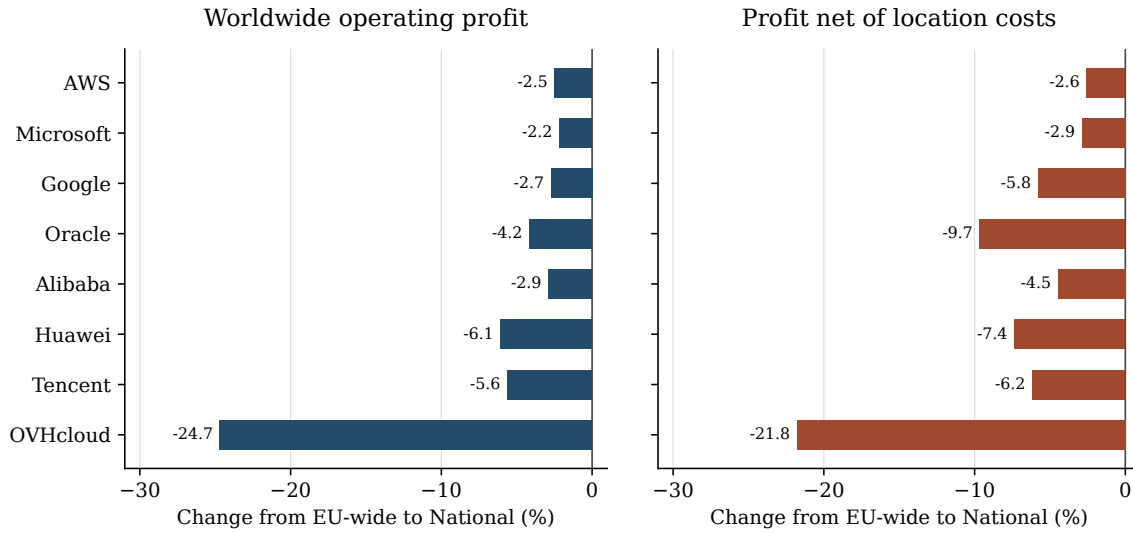
The central equilibria use $\phi = 0$. At $\phi = 1$ we reprice either their policy-specific portfolios or common baseline portfolios, as distinguished in Table 17; neither calculation reoptimizes portfolios under exposure. The baseline reference is common because $d = 0$.

At the central policy portfolios, residual exposure preserves the price-effect signs but reverses the Chinese providers' combined share-change signs. These are conditional repricing results; assessing network responses under exposed residual supply would require reoptimizing portfolios.

F.5 Worldwide provider profits

Figure 7 reports both profit measures. OVHcloud has the largest proportional EU losses: 24.69 percent in operating profit and 21.78 percent in profit net of location costs. Each EU operating margin, the ratio of expected operating profit to expected revenue, changes by less than 0.07 percentage points. Stable margins can coexist with losses of revenue and profit. China profit attribution depends on the maintained partner arrangement; operating margins are not reported there. Share adding-up, core totals, and the profit-cost identity are checked in aggregation, separately from equilibrium verification.

(a) EU: National relative to EU-wide



(b) China: changes in offshore delivery costs

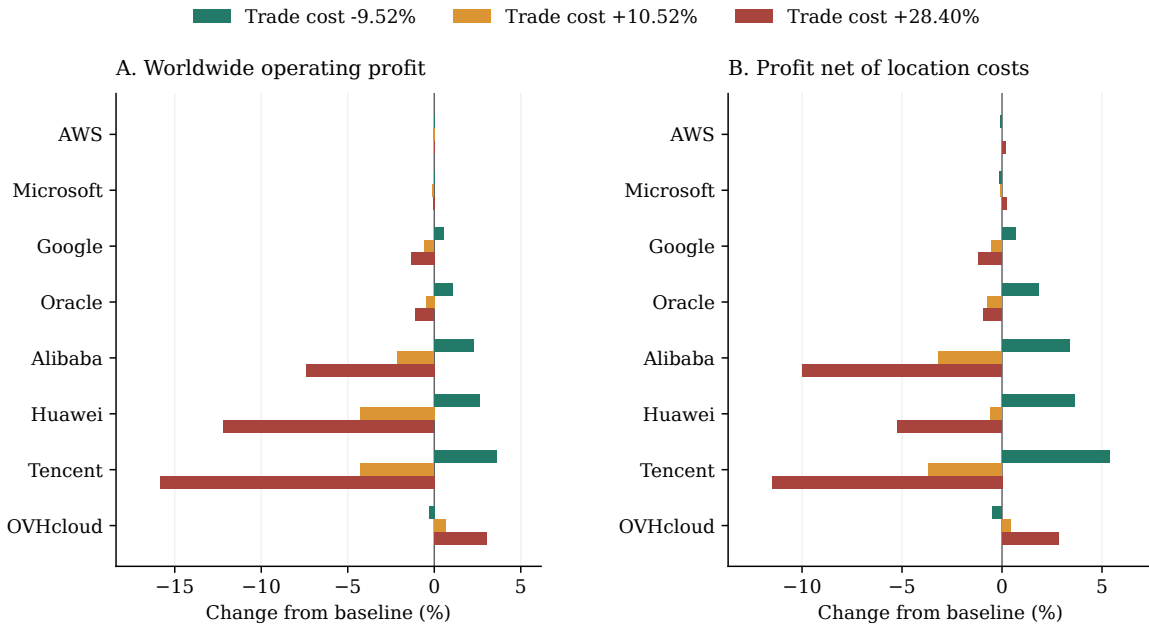


Figure 7: Worldwide provider profits under the two policies

Notes: Ratios use expected profit levels across two integration samples and positive reference denominators. Operating profit subtracts variable production costs; profit net of location costs also subtracts signed core costs, including private draws, and is not accounting net income. China uses unexposed residual supply and attributes all branded partner-path operating profit to AWS and Microsoft.

Table 18: US incidence of the EU and China experiments

Experiment	US customer price (%)	US firms' global profit (%)	US firms' foreign cores (presences)	Cores in the US (presences)
EU: National versus EU-wide	+0.18	-2.48	-0.97	-0.02
China: trade cost -9.52%	-0.08	+0.07	+0.23	+0.02
China: trade cost +10.52%	+0.08	-0.14	-0.17	-0.03
China: trade cost +28.40%	+0.24	-0.20	-0.46	-0.10

Notes: Equilibrium portfolio responses, using the same saved distributions and outcome draws as the main results. US firms are AWS, Microsoft, Google, and Oracle. Global profit is their combined operating profit before location costs; percentage changes compare sums of expected levels, rather than averaging firm growth rates. Foreign cores are their expected core-country presences outside the US. Cores in the US sum the inclusion probabilities of all eight firms. Counts exclude fixed local facilities and partner paths and do not measure capacity. China retains the baseline operating arrangements and unexposed residual supply. Profits follow the model's provider attribution, including its treatment of partner services; they are not US domestic income or a welfare measure. The two applications retain their respective baselines; this table does not evaluate a US policy change.

F.6 US firms and customers

The US incidence comparison aggregates AWS, Microsoft, Google, and Oracle profits and core presences, and separately reports the US customer price index. Table 18 distinguishes these outcomes from cores physically located in the United States.

National rather than EU-wide eligibility lowers these four providers' combined worldwide operating profit by 2.48 percent and reduces their expected core presences outside the United States by 0.97. The US customer price index rises by 0.18 percent. The larger China tightening raises that index by 0.24 percent while lowering the same providers' combined operating profit by 0.20 percent. These are consequences of the two foreign market-access experiments, not estimates of a US policy change or of US welfare. They illustrate why provider nationality, the location of production, and the market served must be distinguished.

G. FURTHER PORTFOLIO-UNIQUENESS CONDITIONS

Containment and localized contraction. Proposition A.1 also applies on a region containing every equilibrium. Invariance gives convergence within it; finite-step containment $\mathcal{B}^L(Q) \subseteq \mathcal{U}$ gives global convergence. These conditions and the pairwise response bound must hold at the specified noise level; a Jacobian evaluated at one equilibrium does not suffice.

G.1 Market effects and signed strategic responses

Different portfolios may have similar market effects. This condition bounds strategic feedback through those effects, retaining all portfolio choices and their Gaussian costs while approximating only the interaction between firms.

Exact market information. Let $\Phi_f(S)$ be a fixed finite vector of market features and let $F_f = \mathbb{E}_{Q_f} \Phi_f(S)$, $F = (F_f)_f$. Require the exact representation

$$V_f(S; Q_{-f}) = d_f(S) + \langle v_f(F_{-f}), \Phi_f(S) \rangle + c_f(F_{-f}), \quad (\text{G.1})$$

where d_f is fixed and c_f does not depend on the own portfolio. For market-additive operating profits and independent firms, the vector can contain indicators for every attainable score in each market. Then F_f records the entire marginal score distributions. Expected scores or coarse bins alone need not suffice. The attainable set $\mathcal{F}_f = \text{conv}\{\Phi_f(S) : S \in \mathcal{X}_f\}$ retains the fact that one portfolio supplies all markets.

Let $\mathcal{P}_f(a)$ denote the portfolio law maximizing $d_f(S) + \langle a, \Phi_f(S) \rangle$ minus the original private costs, and set $R_f(a) = \mathbb{E}_{\mathcal{P}_f(a)} \Phi_f(S)$. Almost-sure uniqueness of this choice implies that fixed points of $F_f = R_f(v_f(F_{-f}))$ correspond one-to-one to portfolio equilibria: their portfolio laws are $\mathcal{P}_f(v_f(F_{-f}))$. Thus equality of equilibrium market features is enough even when Φ_f does not distinguish every portfolio.

Gaussian boundary response. Suppose the cost shocks ε_{fj} are independent standard normal. Consider expected optimized profit as a function of the returns to market features:

$$\mathcal{W}_f(a) = \mathbb{E} \max_{S \in \mathcal{X}_f} \left\{ d_f(S) + \langle a, \Phi_f(S) \rangle - \sigma_\varepsilon \sum_{j \in S} \ell_{fj} \varepsilon_{fj} \right\}.$$

It is convex as an expectation of maxima of affine functions. The finite choice set and almost-sure uniqueness give $\nabla \mathcal{W}_f(a) = R_f(a)$: the marginal profit gain from raising a feature's return equals the expected amount of that feature in the chosen portfolio. Where the response derivative exists it is therefore positive semidefinite. The boundary formula below describes that response; the segment regularity required for the proof is stated separately. At regular payoff values,

$$L_f(a) = DR_f(a) = \sum_e \omega_{fe}(a) u_{fe} u'_{fe}, \quad u_{fe} = \Phi_f(S_e) - \Phi_f(T_e), \quad \omega_{fe} \geq 0. \quad (\text{G.2})$$

Here e ranges over unordered shared boundaries of optimal choice regions. The coordinate boundary budget in (A.6) gives

$$\sum_e \omega_{fe}(a) \leq \beta_f, \quad \beta_f = \frac{1}{\sigma_\varepsilon \sqrt{2\pi}} \sum_{j \in \mathcal{J}_f} \ell_{fj}^{-1}. \quad (\text{G.3})$$

Indeed, $\omega_{fe} = w_e/\sigma_\varepsilon$ in that proof, and every nontrivial boundary changes at least one coordinate. Summing the coordinate budgets bounds the total weight. These statements also hold for averages along payoff segments. The matrix is positive semidefinite, but the interaction between firms below can have either sign.

Retained effects and approximation error. Let $\mathcal{U} \subseteq \prod_f \mathcal{F}_f$ be convex and contain the market features of every equilibrium. Assume $v = (v_f)_f$ is continuously differentiable on a relative neighborhood of $\mathcal{U}$, and that each R_f is absolutely continuous along the connecting payoff segments, with derivative (G.2) almost everywhere. Write $K(F) = Dv(F)$; its own-firm blocks vanish. Fix orthogonal projections P_f onto retained feature spaces, set $P = \text{diag}(P_f)$, and choose an interaction approximation $\tilde{K}(F) = P\tilde{K}(F)P$ with zero diagonal blocks. These spaces remain fixed throughout $\mathcal{U}$.

Choose fixed positive semidefinite matrices A_f on the retained spaces such that

$$P_f L_f(a) P_f \preceq A_f \quad (\text{G.4})$$

throughout every segment joining $v_f(F_{-f})$ and $v_f(F'_{-f})$, for $F, F' \in \mathcal{U}$. This is a population response bound over all feasible portfolios. A matrix evaluated at one equilibrium is insufficient. For $E_{fg}(F) = K_{fg}(F) - \tilde{K}_{fg}(F)$, suppose uniformly on $\mathcal{U}$ and over all feasible portfolio pairs that

$$|\langle \Phi_f(S) - \Phi_f(T), E_{fg}(F)[\Phi_g(U) - \Phi_g(V)] \rangle| \leq \gamma_{fg}. \quad (\text{G.5})$$

This condition includes all omitted input and output directions.

Put $A = \text{diag}(A_f)$ and fix positive firm weights $\Lambda = \text{diag}(\lambda_f I)$. With $\text{Sym}(B) = (B + B')/2$, choose a nonnegative uniform upper h_B :

$$\lambda_{\max} \left[A^{1/2} \text{Sym} \left(\Lambda^{1/2} \tilde{K}(F) \Lambda^{-1/2} \right) A^{1/2} \right] \leq h_B, \quad F \in \mathcal{U}. \quad (\text{G.6})$$

Define $\delta_{fg} = \sqrt{\beta_f \beta_g} \gamma_{fg}$ and the symmetric nonnegative matrix D by

$$D_{ff} = 0, \quad D_{fg} = \frac{1}{2} \left[\sqrt{\frac{\lambda_f}{\lambda_g}} \delta_{fg} + \sqrt{\frac{\lambda_g}{\lambda_f}} \delta_{gf} \right]. \quad (\text{G.7})$$

Proposition G.1 (Uniqueness from market effects). *Under the preceding assumptions and those of Proposition 2, the portfolio BNE is unique if*

$$\underbrace{h_B}_{\text{retained strategic effects}} + \underbrace{\rho(D)}_{\text{approximation allowance}} < 1. \quad (\text{G.8})$$

Proof. Suppose there are two equilibria. Their difference in market features must reproduce itself through the change in returns and firms' choices. Integrating separately along the payoff and market-feature segments gives

$$\Delta F = \bar{L}\Delta v, \quad \Delta v = \bar{K}\Delta F.$$

The two averages need not use the same path. If $\Delta F \neq 0$, then $x = \bar{L}^{-1/2}\Delta v \neq 0$ satisfies $x = \bar{L}^{-1/2}\bar{K}\bar{L}^{-1/2}x$. Firm weights commute with the block-diagonal $\bar{L}$, so $y = \Lambda^{1/2}x$ obeys the corresponding weighted equation.

We bound the feedback along this difference by separating retained market effects from the omitted interactions. For the retained part, $P\bar{L}P \leq A$ implies $P\bar{L}^{-1/2} = A^{1/2}C$ for a contraction C on the appropriate ranges. Consequently its symmetric quadratic form is bounded above by $h_B\|y\|^2$, using (G.6) and $h_B \geq 0$. This argument allows singular response matrices and uses no inverse of $\bar{L}$ or A .

For the remainder, the boundary representation of the segment averages and (G.5) imply

$$\|\bar{L}_f^{-1/2}E_{fg}(F)\bar{L}_g^{-1/2}\|_{\text{op}}^2 \leq \text{tr}(\bar{L}_f E_{fg}(F)\bar{L}_g E_{fg}(F)') \leq \beta_f \beta_g \gamma_{fg}^2.$$

The same norm bound holds after averaging E on the market-feature segment. Block norm comparison bounds the symmetric weighted remainder by $\rho(D)$. Adding the two parts gives $\|y\|^2 \leq [h_B + \rho(D)]\|y\|^2$, a contradiction. Thus $\Delta F = 0$ and the unique response to these features gives identical portfolio laws. Existence follows from Proposition 2. $\square$

The error condition has an economic interpretation. When the exact features record complete market score laws, a contrast in (G.5) is a four-corner difference of repriced profits. A differentiable, finite separable approximation with uniform profit error ϵ_{fgn}^π in market n therefore gives $\gamma_{fg} \leq 4 \sum_n \epsilon_{fgn}^\pi$, provided its induced interaction operator lies in the fixed retained spaces. The bound must cover all relevant other-rival laws. Approximation compresses market interactions; it does not discard portfolios or truncate private-cost tails.

Condition (G.8) can retain cancellations lost by a nonnegative total-variation comparison. It does not generally dominate Theorem 1, because it uses different bounds and additional uniformity requirements. It establishes uniqueness, not convergence of a particular response iteration. The calibrated applications do not verify its population response bounds or its uniform signed margin.